

Codebook

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1 basic

1.1 build

```

1 code tem.cpp
2 ...
3 for i in {A..M}; do cp tem.cpp \${i}.cpp; done;
4 code *.cpp
5 code run
6
7 寫入以下編譯與執行指令至 run 檔案中：
8 g++ -std=c++20 -DABS -Wall -Wextra -Wshadow $1.cpp -o $1 && ./
9   $1
10 chmod +x run

```

1.2 default

```

1 #include <bits/stdc++.h>
2 #define int long long
3 #define pii pair<int, int>
4 #define F first
5 #define S second
6 #define pb push_back
7 #define all(x) x.begin(), x.end()
8 #define sz(x) (int)x.size()
9 #define FOR(i, a, b) for(int i = a; i <= b; ++i)
10 #define IO ios::sync_with_stdio(0), cin.tie(0)
11 using namespace std;
12 void debug() { cerr << endl; }
13 template <typename T, typename ...U>
14 void debug(T i, U ...j) { cerr << i << ' ', debug(j...); }
15 #define test(...) debug("[ " #__VA_ARGS__ " ]:", __VA_ARGS__)
16
17 signed main() {
18     IO;
19
20 }

```

1.3 division

```

1 ll floor(ll a, ll b) {
2     return a / b - (a < 0 && a % b);
3 }
4 ll ceil(ll a, ll b) {
5     return a / b + (a > 0 && a % b);
6 }
7 a / b < x -> floor(a, b) + 1 <= x
8 a / b <= x -> ceil(a, b) <= x
9 x < a / b -> x <= ceil(a, b) - 1
10 x <= a / b -> x <= floor(a, b)

```

1.4 fast

```

1 #pragma GCC optimize("Ofast,inline,unroll-loops")
2 #pragma GCC target("bmi,bmi2,lzcnt,popcnt,avx2")
3 #define windows_system
4 #ifdef windows_system
5     /* For Windows */
6 inline int getchar_unlocked() { return getchar(); }
7 inline void putchar_unlocked(char _c) { putchar(_c); }
8 #endif
9 inline int in() {
10     int re = 0;
11     char c = getchar_unlocked();
12     while (c == ' ' || c == '\n') c = getchar_unlocked();
13     while (c >= '0' && c <= '9') re = (re << 3) + (re << 1) + c - '0', c = getchar_unlocked();
14     return re;
15 }
16 inline void out(int x) {
17     char str[20];
18     int pos = 0;
19     do {
20         str[pos++] = x % 10 + '0';
21         x /= 10;
22     } while (x);
23     for (int i = pos - 1; i >= 0; i--) putchar_unlocked(str[i]);
24     putchar_unlocked('\n');
25 }

```

1.5 pragma

```

1 #pragma GCC optimize("Ofast,no-stack-protector")
2 #pragma GCC optimize("no-math-errno,unroll-loops")
3 #pragma GCC target("sse,sse2,sse3,ssse3,sse4")
4 #pragma GCC target("popcnt,abm,mmx,avx,arch=skylake")

```

1.6 random

```

1 mt19937 seed(chrono::steady_clock::now().time_since_epoch().
2   count());
3 inline int rnd(int l, int r) {
4     return uniform_int_distribution<int>(l, r)(seed);
5 }
6 inline double drnd(double l, double r) {
7     return uniform_real_distribution<double>(l, r)(seed);
8 }

```

2 data structure

2.1 2D Segment Tree

```

1 struct Node2 {
2     int val; Node2 *lc, *rc;
3     Node2 () {
4         val = 0;
5         lc = rc = nullptr;
6     }
7 };
8 struct Node1 {
9     Node2 *c; Node1 *lc, *rc;
10    Node1 () {
11        c = nullptr;
12        lc = rc = nullptr;
13    }
14 };
15 int N;
16 int Val2(Node2 *n2) { return n2 ? n2->val : 0; }
17 void pull2(Node2 *n2) { n2->val = Val2(n2->lc) + Val2(n2->rc); }
18 void update2(Node2 *n2, int l, int r, int id, int v) {
19     if (l == r) {
20         n2->val += v; return;
21     }
22     int mid = (l + r) >> 1;
23     if (id <= mid) {
24         if (!n2->lc) n2->lc = new Node2();
25         update2(n2->lc, l, mid, id, v);
26     } else {
27         if (!n2->rc) n2->rc = new Node2();
28         update2(n2->rc, mid + 1, r, id, v);
29     }
30     pull2(n2);
31 }
32 void update1(Node1 *n1, int l, int r, int qx, int qy, int v) {
33     if (!n1->c) n1->c = new Node2();
34     update2(n1->c, l, N, qx, v);
35     if (l == r) return;
36     int mid = (l + r) >> 1;
37     if (qx <= mid) {
38         if (!n1->lc) n1->lc = new Node1();
39         update1(n1->lc, l, mid, qx, qy, v);
40     } else {
41         if (!n1->rc) n1->rc = new Node1();
42         update1(n1->rc, mid + 1, r, qx, qy, v);
43     }
44 }
45 int query2(Node2 *n2, int l, int r, int ql, int qr) {
46     if (!n2) return 0;
47     if (ql <= l && r <= qr) return n2->val;
48     int mid = (l + r) >> 1;
49     if (qr <= mid)
50         return query2(n2->lc, l, mid, ql, qr);
51     else if (mid + 1 <= ql)
52         return query2(n2->rc, mid + 1, r, ql, qr);
53     else
54         return query2(n2->lc, l, mid, ql, qr) + query2(n2->rc, mid + 1, r, ql, qr);
55 }
56 int query1(Node1 *n1, int l, int r, int qxl, int qxr, int qyl,
57   int qyr) {
58     if (!n1) return 0;
59     if (qxl <= l && r <= qxr) return query2(n1->c, l, N, qyl, qyr);
60     int mid = (l + r) >> 1;
61     if (qxr <= mid)
62         return query1(n1->lc, l, mid, qxl, qxr, qyl, qyr);
63     else if (mid + 1 <= qxl)

```

```

63     return query1(n1->rc, mid + 1, r, qxl, qxr, qyl, qyr);
64 else
65     return query1(n1->lc, l, mid, qxl, qxr, qyl, qyr) + query1(
66         n1->rc, mid + 1, r, qxl, qxr, qyl, qyr);

```

2.2 Binary Indexed Tree

```

1 struct BIT {
2     int N;
3     vector<int> info;
4     BIT(int n) : N(n), info(n + 1) {}
5     void update(int idx, int val) {
6         for (int i = idx; i <= N; i += i & -i)
7             info[i] += val;
8     }
9     int query(int idx) {
10        int res = 0;
11        for (int i = idx; i > 0; i -= i & -i)
12            res += info[i];
13        return res;
14    }
15    int kth(int k) { // kth small
16        int res = 0;
17        for (int i = 1 << __lg(N); i > 0; i >= 1) {
18            if (res + i <= N && info[res + i] < k) {
19                res += i;
20                k -= info[res];
21            }
22        }
23        return res + 1;
24    }
25 };

```

2.3 bitset

```

1 #pragma GCC target("popcnt")
2 // 15 = 1111
3 bitset<10> bit(15); // 宣告長度 = 10 的 bitset
4 cout << bit.count() << "\n"; // 回傳幾個 1
5 cout << bit.none() << "\n"; // 回傳是否全為 0
6 cout << bit.any() << "\n"; // 回傳是否有 1
7 cout << bit._Find_first() << "\n"; // 回傳第一個 bit = 1 的位
8 // 置
9 int x = 2;
10 cout << bit._Find_next(x) << "\n"; // 回傳 x 後 · 第一個 bit =
11 // 1 的位置

```

2.4 Li Chao Segment Tree

```

1 // MIN LeeChao Segment Tree
2 struct line {
3     int m, k;
4     line *lc, *rc;
5     line() { m = 0, k = INT_MAX, lc = nullptr, rc = nullptr; }
6     line(int _m, int _k) { m = _m, k = _k, lc = nullptr, rc =
7         nullptr; }
8     int calc(int x) {
9         return m * x + k;
10    }
11    void push(int l, int r) {
12        if (l == r) return;
13        if (!lc) lc = new line();
14        if (!rc) rc = new line();
15    }
16 };
17 struct LeeChao {
18     line* root;
19     LeeChao() : root(new line()) {}
20     void update(line* node, int l, int r, line add) {
21         if (l == r) {
22             if (add.calc(l) < node->calc(l)) {
23                 swap(add.m, node->m);
24                 swap(add.k, node->k);
25             }
26             return;
27         }
28         // 預設 seg[node] 的斜率 < add 的斜率
29         if (node->m > add.m) {
30             swap(add.m, node->m);
31             swap(add.k, node->k);
32         }
33         node->push(l, r);
34         int mid = (l + r) >> 1;
35         // 將小的 f(x) 留下
36         if (node->calc(mid) > add.calc(mid)) {

```

```

36         swap(add.m, node->m);
37         swap(add.k, node->k);
38         update(node->rc, mid + 1, r, add);
39     } else {
40         update(node->lc, l, mid, add);
41     }
42 }
43 int query(line* node, int l, int r, int x) {
44     if (node->k == INT_MAX) return INT_MAX;
45     if (l == r) return node->calc(x);
46     int mid = (l + r) >> 1;
47     if (x <= mid)
48         return min(node->calc(x), query(node->lc, l, mid, x));
49     else
50         return min(node->calc(x), query(node->rc, mid + 1, r, x));
51 }
52 void update(line l) { update(root, 1, N, l); }
53 int query(int x) { return query(root, 1, N, x); }
54 };

```

2.5 pbds

```

1 #include <ext/pb_ds/assoc_container.hpp>
2 #include <ext/pb_ds/tree_policy.hpp>
3 using namespace __gnu_pbds;
4 void solve() {
5     tree<int, null_type, greater<int>, rb_tree_tag,
6         tree_order_statistics_node_update> bst;
7     bst.insert(1);
8     int a = *bst.find_by_order(2); // 第 k + 1 大的數
9     int b = bst.order_of_key(2); // 有幾個數 > k
10    // -----
11    tree<int, null_type, less<int>, rb_tree_tag,
12        tree_order_statistics_node_update> bst;
13    bst.insert(1);
14    int a = *bst.find_by_order(2); // 第 k + 1 小的數
15    int b = bst.order_of_key(2); // 有幾個數 < k
16 }

```

2.6 Range Set

```

1 struct RangeSet { // [L, r)
2     set<pii> st;
3     void cut(int x) {
4         auto it = st.lower_bound({x + 1, -1});
5         if (it == st.begin()) return;
6         auto [l, r] = *prev(it);
7         if (l >= x || x >= r) return;
8         st.erase(prev(it));
9         st.insert({l, x});
10        st.insert({x, r});
11    }
12    vector<pii> split(int l, int r) {
13        // remove and return ranges in [L, r)
14        cut(l), cut(r);
15        vector<pii> res;
16        while (true) {
17            auto it = st.lower_bound({l, -1});
18            if (it == st.end() || r <= it->first) break;
19            res.push_back(*it), st.erase(it);
20        }
21        return res;
22    }
23    void insert(int l, int r) {
24        // add a range [L, r), [L, r) not in st
25        auto it = st.lower_bound({l, r});
26        if (it != st.begin() && prev(it)->second == l)
27            l = prev(it)->first, st.erase(prev(it));
28        if (it != st.end() && r == it->first)
29            r = it->second, st.erase(it);
30        st.insert({l, r});
31    }
32    bool count(int x) {
33        auto it = st.lower_bound({x + 1, -1});
34        return it != st.begin() && prev(it)->first <= x && x < prev(
35            it)->second;
36    };

```

2.7 Sparse Table

```

1 int a[N];
2 struct Sparse_table {
3     int n;
4     vector<vector<int>> st;

```

```

5 Sparse_table(int _n) : n(_n), st(_n + 1, vector<int>(__lg(_n)
6     + 1)) {
7     for (int i = 1; i <= n; i++) st[i][0] = a[i];
8     for (int lg = 1; lg <= __lg(n); lg++) {
9         int len = 1 << (lg - 1);
10        for (int i = 1; i + len <= n; i++)
11            st[i][lg] = max(st[i][lg - 1], st[i + len][lg - 1]);
12    }
13    int query(int l, int r) {
14        int lg = __lg(r - l + 1);
15        int len = 1 << lg;
16        return max(st[l][lg], st[r - len + 1][lg]);
17    }
18 };

```

2.8 Standard Template Library

```

1 struct Data {
2     int x, y;
3 };
4 // priority_queue
5 struct cmp {
6     bool operator()(Data a, Data b) {
7         return a.x < b.x;
8     }
9 };
10 priority_queue<Data, vector<Data>, cmp> pq;
11 // set
12 struct cmp {
13     bool operator() (Data p, Data q) {
14         return p.x < q.x;
15     }
16 };
17 set<Data, cmp> st;

```

2.9 Treap

```

1 struct node {
2     int data, sz;
3     node *l, *r;
4     node(int k) : data(k), sz(1), l(0), r(0) {}
5     void up() {
6         sz = 1;
7         if (l) sz += l->sz;
8         if (r) sz += r->sz;
9     }
10    void down() {}
11 };
12 int sz(node *a) { return a ? a->sz : 0; }
13 node *merge(node *a, node *b) {
14     if (!a || !b) return a ? a : b;
15     if (rand() % (sz(a) + sz(b)) < sz(a))
16         return a->down(), a->r = merge(a->r, b), a->up(), a;
17     return b->down(), b->l = merge(a, b->l), b->up(), b;
18 }
19 void split1(node *o, node *&a, node *&b, int k) {
20     if (!o) return a = b = 0, void();
21     o->down();
22     if (o->data <= k)
23         a = o, split1(o->r, a->r, b, k), a->up();
24     else
25         b = o, split1(o->l, a, b->l, k), b->up();
26 }
27 void split2(node *o, node *&a, node *&b, int k) {
28     if (sz(o) <= k) return a = o, b = 0, void();
29     o->down();
30     if (sz(o->l) + 1 <= k)
31         a = o, split2(o->r, a->r, b, k - sz(o->l) - 1);
32     else
33         b = o, split2(o->l, a, b->l, k);
34     o->up();
35 }
36 node *kth(node *o, int k) {
37     if (k <= sz(o->l)) return kth(o->l, k);
38     if (k == sz(o->l) + 1) return o;
39     return kth(o->r, k - sz(o->l) - 1);
40 }
41 int Rank(node *o, int key) {
42     if (!o) return 0;
43     if (o->data < key)
44         return sz(o->l) + 1 + Rank(o->r, key);
45     else
46         return Rank(o->l, key);
47 }
48 bool erase(node *&o, int k) {
49     if (!o) return 0;
50     if (o->data == k) {
51         node *t = o;
52         o->down(), o = merge(o->l, o->r);
53         delete t;

```

```

54     return 1;
55 }
56 node *t = k < o->data ? o->l : o->r;
57 return erase(t, k) ? o->up(), 1 : 0;
58 }
59 void insert(node *&o, int k) {
60     node *a, *b;
61     split(o, a, b, k),
62     o = merge(a, merge(new node(k), b));
63 }
64 void interval(node *&o, int l, int r) {
65     node *a, *b, *c;
66     split2(o, a, b, l - 1), split2(b, b, c, r);
67     // operate
68     o = merge(a, merge(b, c));
69 }

```

2.10 Undo Disjoint Set

```

1 struct DSU {
2     vector<int> parent, size, tag;
3     vector<tuple<int, int, int, int, int>> history;
4     DSU(int n) {
5         parent.resize(n + 1);
6         size.resize(n + 1, 1);
7         tag.resize(n + 1, 0);
8         iota(all(parent), 0);
9     }
10    int find(int x) {
11        if (parent[x] == x) return x;
12        return find(parent[x]);
13    }
14    bool merge(int x, int y) {
15        x = find(x), y = find(y);
16        if (x == y) return false;
17        if (size[x] < size[y]) swap(x, y);
18        history.pb({x, y, parent[y], tag[x], size[x]});
19        parent[y] = x;
20        size[x] += size[y];
21        return true;
22    }
23    void add_tag(int head, int val) {
24        tag[head] += val;
25    }
26    void undo() {
27        auto [px, py, parent_py, base, sz_x] = history.back();
28        parent[py] = parent_py;
29        size[px] = sz_x;
30        int delta = tag[px] - base;
31        tag[py] += delta;
32        history.pop_back();
33    }
34 };

```

2.11 Unorder Map

```

1 struct custom_hash {
2     static uint64_t splitmix64(uint64_t x) {
3         x += 0x9e3779b97f4a7c15;
4         x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9;
5         x = (x ^ (x >> 27)) * 0x94d049bb133111eb;
6         return x ^ (x >> 31);
7     }
8     size_t operator()(uint64_t a) const {
9         static const uint64_t FIXED_RANDOM = chrono::steady_clock::
10            now().time_since_epoch().count();
11         return splitmix64(i + FIXED_RANDOM);
12     }
13 };
14 unordered_map<int, int, custom_hash> m1;

```

3 flow matching

3.1 Dinic

```

1 struct Flow {
2     struct Edge {
3         // rc : residual capacity
4         int u, rc, rev;
5     };
6     vector<vector<Edge>> G;
7     vector<int> dis, it;
8     Flow(int n) : G(n), dis(n), it(n) {}
9     void add(int v, int u, int c) {
10         G[v].push_back({u, c, sz[G[u]]});
11         G[u].push_back({v, 0, sz[G[v]} - 1});
12     }

```

```

13 int dfs(int v, int t, int f) {
14     if (v == t || f == 0) return f;
15     for (int &i = it[v]; i < sz(G[v]); i++) {
16         auto &[u, rc, rev] = G[v][i];
17         if (dis[u] != dis[v] + 1) continue;
18         int res = dfs(u, t, min(f, rc));
19         if (res <= 0) continue;
20         rc -= res;
21         G[u][rev].rc += res;
22         return res;
23     }
24     return 0;
25 }
26 int flow(int s, int t) {
27     int ans = 0;
28     for (int l = 30; l >= 0; l--) while (true) {
29         fill(all(dis), INT_MAX);
30         queue<int> Q;
31         Q.push(s);
32         dis[s] = 0;
33         while (!Q.empty()) {
34             int v = Q.front(); Q.pop();
35             for (auto [u, rc, rev] : G[v]) {
36                 if ((rc >> 1) <= 0 || dis[u] < INT_MAX) continue;
37                 dis[u] = dis[v] + 1;
38                 Q.push(u);
39             }
40         }
41         if (dis[t] == INT_MAX) break;
42
43         fill(all(it), 0);
44         while (true) {
45             int res = dfs(s, t, INT_MAX);
46             if (res <= 0) break;
47             ans += res;
48         }
49     }
50     return ans;
51 }
52 };

```

3.2 flow building

Dinic

基本建邊

- 有向邊 $u \rightarrow v$ · 容量 c : `add(u, v, c)` · 反向邊容量為 0。
- 無向邊 $u - v$ · 容量 c : `add(u, v, c)` · 並把反向邊容量也設為 c · 不需另外再加一條邊。
- 多源點或多匯點：開超級源 S · 連 $S \rightarrow$ 每個源點 (∞) ；開超級匯 T · 連每個匯點 $\rightarrow T(\infty)$ 。
- 點有容量 c ：拆點為 $v_{in} \rightarrow v_{out}(c)$ · 進入的邊接到 v_{in} · 出去的邊從 v_{out} 出發。

匹配與分配

- 二分圖匹配： $S \rightarrow L(1)$ 、 $L \rightarrow R(1)$ 、 $R \rightarrow T(1)$ · 最大流即最大匹配數。
- 每個點最多配 k 個：把 S 側或 T 側的邊容量改為 k 。
- 同一對物品只能配一次：中間的邊容量設為 1。
- 多對多分配： $S \rightarrow$ 人 (可做件數)、人 $\rightarrow$ 工作 (1 或上限)、工作 $\rightarrow T$ (需求量)；是否可行看是否滿流。

最小割模型

- 題目出現「代價」「二選一」「放棄」等字眼時 · 考慮最小割。
- 某條邊不能被割：容量設為 ∞ 。
- 最大權閉合子圖 (選 i 就必須選 j)：
 - 收益 $p > 0$: $S \rightarrow i(p)$
 - 花費 c : $i \rightarrow T(c)$
 - 依賴關係： $i \rightarrow j(\infty)$
 - 答案 = 正收益總和 - 最大流
- 二選一 (每點選 A 或 B · 相鄰兩點選得不同要付罰金 w)：
 - 規定點在 S 側代表選 A · 在 T 側代表選 B 。
 - $S \rightarrow i$ (選 B 的代價) · $i \rightarrow T$ (選 A 的代價)

- 兩點選得不同的罰金： $i \rightarrow j(w)$ 與 $j \rightarrow i(w)$
- 答案 = 最小割

- 強制某點一定選 A 或一定選 B : $S \rightarrow i(\infty)$ 或 $i \rightarrow T(\infty)$ 。
- 網絡或棋盤類：先黑白染色成二分圖。
 - 最小點覆蓋 = 最大匹配
 - 最大獨立集 = 總點數 - 最大匹配

路徑類

- 邊不相交的路徑數：每條邊容量 1 · 直接跑最大流。
- 點不相交的路徑數：拆點 · $v_{in} \rightarrow v_{out}(1)$ 。
- DAG 最小路徑覆蓋：
 - 每點拆成出點與入點： $S \rightarrow u_{out}(1)$ 、 $u_{in} \rightarrow T(1)$
 - 原圖的邊 $u \rightarrow v$: $u_{out} \rightarrow v_{in}(1)$
 - 答案 = n - 最大流
 - 若路徑可重複經過點 · 先做遞移閉包再建圖。
- 與時間有關 (每單位時間走一步、各時刻容量不同)：建分層圖 · 把每個點依時間複製成 (v, t) · 邊從第 t 層連到第 $t+1$ 層。

有上下界的流 (邊 $u \rightarrow v$ 的流量需在 $[l, r]$ 內)

- 建 $u \rightarrow v(r-l)$ · 並記錄 $in[v] += l$ 、 $out[u] += l$ 。
- 另開源匯 S' 、 T' ：
 - 若 $in[x] - out[x] > 0$: $S' \rightarrow x$ (差值)
 - 若差值 < 0 : $x \rightarrow T'$ (-差值)
- 若題目本身有源匯 s, t · 再加 $t \rightarrow s(\infty)$ 。
- 從 S' 到 T' 跑滿流才代表有可行解。
- 求最大流：接著在殘量網絡上跑 $s \rightarrow t$ 。
- 求最小流：接著在殘量網絡上跑 $t \rightarrow s$ · 再用原流量減去。

MCMF

基本建邊

- 每條邊為 `add(u, v, cap, cost)` · 反向邊容量 0 · 費用 $-cost$ 。

常見情境

- 帶權二分匹配 (指派問題) : $S \rightarrow L(1, 0)$ 、 $L \rightarrow R(1, w)$ 、 $R \rightarrow T(1, 0)$ 。
- 求最大收益：費用取負 $cost = -w$ · 最後答案再取負。
- 只求收益最大、不要求流量最大：
 - 做法一：增廣時一旦最短路費用 ≥ 0 就停止。
 - 做法二：每個點加一條 $\rightarrow T(1, 0)$ 的邊 · 讓它可以選擇不配。
- 指定總流量為 F ：開新源 S_0 · 連 $S_0 \rightarrow S(F, 0)$ 。
- 點有費用或收益：拆點 $v_{in} \rightarrow v_{out}$ (容量, 費用)；例如每點只能拿一次收益 p · 建 $(1, -p)$ 。
- 點可經過多次 · 但收益只能拿一次：拆點並放兩條平行邊 $(1, -p)$ 與 $(\infty, 0)$ 。
- 邊際成本遞增 (例如費用為 x^2)：拆成多條容量 1 的平行邊 · 費用依序為 $1, 3, 5, \dots$ (即 x^2 的差分)。前提是費用為凸 · 一條比一條貴 · 才會依序被選用。
- k 條路徑的最大權 (每格只能取一次、共走 k 次)：格子拆點並放兩條平行邊 $(1, -val)$ 與 $(k-1, 0)$ · 再連 $S \rightarrow$ 起點 $(k, 0)$ 。
- 區間選擇 (選帶權區間 · 每個位置最多被 k 個區間覆蓋)：
 - 座標離散化後建一條鏈： $i \rightarrow i+1(k, 0)$ 。
 - 每個區間 $[l, r]$: $l \rightarrow r(1, -w)$ 。
 - 從鏈頭到鏈尾跑流量 k 。
- 資源用完後隔一段時間可再用 (餐巾問題類型)：
 - 按天拆成「用完」與「需要」兩類點。
 - $S \rightarrow$ 第 i 天用完 (用量, 0) · 第 i 天需要 $\rightarrow T$ (需求, 0)。
 - 買新的： $S \rightarrow$ 需要 $(\infty, 買價)$

- 快速回收：用完 $i \rightarrow$ 需要 $i + a$ (∞ , 快速費用)
- 慢速回收：用完 $i \rightarrow$ 需要 $i + b$ (∞ , 慢速費用)
- 留到隔天：用完 $i \rightarrow$ 用完 $i + 1$ ($\infty, 0$)

- 某條邊一定要走：費用設為 $-BIG$ ，或改用上下界流。

注意事項

- Dinic's 算法：時間複雜度 $O(V^2E)$
- MCMF：時間複雜度 $O(FEV)$ ，其中 F 為最大流量。

建模方式對照表

題目關鍵字	建模方式
配對、分配、容量上限	最大流
點不能重複用、點有限制	拆點
最少代價阻止某事、二選一、依賴關係	最小割 (∞ 表示不能切)
配對或運送時帶權重、成本	MCMF
費用隨數量遞增	拆成平行邊，費用遞增
跟時間步有關	分層圖
至少、必須	上下界流

3.3 matching

```

1 const int N = 2005;
2 vector<int> G[N];
3 struct Matching {
4     int n;
5     vector<bool> vis; vector<int> match;
6     Matching(int _n) : n(_n), vis(_n + 1), match(_n + 1, -1) {}
7     bool dfs(int v) {
8         vis[v] = true;
9         for (int u : G[v])
10             if (match[u] == -1 || (! vis[match[u]] && dfs(match[u])))
11                 {
12                     match[u] = v;
13                     return true;
14                 }
15         return false;
16     }
17     int solve() {
18         int cnt = 0;
19         for (int i = 1; i <= n; i++) {
20             vis.assign(n + 1, 0);
21             if (dfs(i)) cnt++;
22         }
23         return cnt;
24     };

```

3.4 matching general

```

1 // O(V^3)
2 // if graph.match[i] == n, node i is not matched
3 // else, i is matched with graph.match[i]
4 struct Matching { // 0-based
5     int fa[N], pre[N], match[N], s[N], v[N], n, tk;
6     vector<int> g[N];
7     queue<int> q;
8     int Find(int u) {
9         return u == fa[u] ? u : fa[u] = Find(fa[u]);
10    }
11    int lca(int x, int y) {
12        tk++;
13        x = Find(x), y = Find(y);
14        for (; ; swap(x, y)) {
15            if (x != n) {
16                if (v[x] == tk) return x;
17                v[x] = tk;
18                x = Find(pre[match[x]]);
19            }
20        }
21    }
22    void blossom(int x, int y, int l) {
23        while (Find(x) != l) {
24            pre[x] = y, y = match[x];
25            if (s[y] == 1) q.push(y), s[y] = 0;
26            if (fa[x] == x) fa[x] = l;
27            if (fa[y] == y) fa[y] = l;
28            x = pre[y];
29        }
30    }
31    bool bfs(int r) {
32        for (int i = 0; i <= n; ++i) fa[i] = i, s[i] = -1;
33        while (!q.empty()) q.pop();
34        q.push(r);
35        s[r] = 0;

```

```

36 while (!q.empty()) {
37     int x = q.front(); q.pop();
38     for (int u : g[x]) {
39         if (s[u] == -1) {
40             pre[u] = x, s[u] = 1;
41             if (match[u] == n) {
42                 for (int a = u, b = x, last; b != n; a = last, b =
43                     pre[a])
44                     last = match[b], match[b] = a, match[a] = b;
45                 return true;
46             }
47             q.push(match[u]);
48             s[match[u]] = 0;
49         } else if (!s[u] && Find(u) != Find(x)) {
50             int l = lca(u, x);
51             blossom(x, u, l);
52             blossom(u, x, l);
53         }
54     }
55     return false;
56 }
57 int solve() {
58     int res = 0;
59     for (int x = 0; x < n; ++x) {
60         if (match[x] == n) res += bfs(x);
61     }
62     return res;
63 }
64 void init(int _n) {
65     n = _n, tk = 0;
66     for (int i = 0; i <= n; ++i) match[i] = pre[i] = n;
67     for (int i = 0; i < n; ++i) g[i].clear(), v[i] = 0;
68 }
69 void add_edge(int u, int v) {
70     g[u].push_back(v), g[v].push_back(u);
71 }
72 };

```

3.5 matching general weight

```

1 // O(V^3)
2 // if graph.match[i] == 0, node i is not matched
3 // else, i is matched with graph.match[i]
4 struct WeightGraph { // 1-based
5     static const int inf = INT_MAX;
6     static const int maxn = 514;
7     struct edge {
8         int u, v, w;
9         edge(){}
10        edge(int u, int v, int w): u(u), v(v), w(w) {}
11    };
12    int n, n_x;
13    edge g[maxn * 2][maxn * 2];
14    int lab[maxn * 2];
15    int match[maxn * 2], slack[maxn * 2], st[maxn * 2], pa[maxn *
16        2];
17    int flo_from[maxn * 2][maxn + 1], S[maxn * 2], vis[maxn * 2];
18    vector<int> flo[maxn * 2];
19    queue<int> q;
20    int e_delta(const edge &e) { return lab[e.u] + lab[e.v] - g[e
21        .u][e.v].w * 2; }
22    void update_slack(int u, int x) { if (!slack[x] || e_delta(g[
23        u][x]) < e_delta(g[slack[x]][x])) slack[x] = u; }
24    void set_slack(int x) {
25        slack[x] = 0;
26        for (int u = 1; u <= n; ++u)
27            if (g[u][x].w > 0 && st[u] != x && S[st[u]] == 0)
28                update_slack(u, x);
29    }
30    void q_push(int x) {
31        if (x <= n) q.push(x);
32        else for (size_t i = 0; i < flo[x].size(); i++) q_push(flo[
33            x][i]);
34    }
35    void set_st(int x, int b) {
36        st[x] = b;
37        if (x > n) for (size_t i = 0; i < flo[x].size(); ++i)
38            set_st(flo[x][i], b);
39    }
40    int get_pr(int b, int xr) {
41        int pr = find(flo[b].begin(), flo[b].end(), xr) - flo[b].
42            begin();
43        if (pr % 2 == 1) {
44            reverse(flo[b].begin() + 1, flo[b].end());
45            return (int)flo[b].size() - pr;
46        }
47        return pr;

```



```

19 while (!Q.empty()) {
20     int v = Q.front(); Q.pop();
21     in_q[v] = false;
22     for (int i = 0; i < (int)G[v].size(); i++) {
23         auto [u, rc, k, rv] = G[v][i];
24         if (rc > 0 && dis[v] + k < dis[u]) {
25             dis[u] = dis[v] + k;
26             par[u] = v;
27             par_eid[u] = i;
28             if (!in_q[u]) Q.push(u);
29             in_q[u] = true;
30         }
31     }
32 }
33 return dis[t];
34 }
35 pair<int, int> flow(int s, int t) {
36     int fl = 0, cost = 0, d;
37     while ((d = spfa(s, t)) < LONG_LONG_MAX) {
38         int cur = LONG_LONG_MAX;
39         for (int v = t; v != s; v = par[v])
40             cur = min(cur, G[par[v]][par_eid[v]].rc);
41         fl += cur;
42         cost += d * cur;
43         for (int v = t; v != s; v = par[v]) {
44             G[par[v]][par_eid[v]].rc -= cur;
45             G[v][G[par[v]][par_eid[v]].rv].rc += cur;
46         }
47     }
48     return {fl, cost};
49 }
50 };

```

4 geometry

4.1 basic

```

1 const double eps = 1e-8, PI = acos(-1);
2 int sign(double x)
3 { return fabs(x) <= eps ? 0 : (x > 0 ? 1 : -1); }
4 double normalize(double x) {
5     while (x < -eps) x += PI * 2;
6     while (x > PI * 2 + eps) x -= PI * 2;
7     return x; }
8 template <typename T> struct P {
9     T x, y;
10    P (T _x = 0, T _y = 0) : x(_x), y(_y) {}
11    P<T> operator + (P<T> o) {
12        return P<T>(x + o.x, y + o.y);}
13    P<T> operator - (P<T> o) {
14        return P<T>(x - o.x, y - o.y);}
15    P<T> operator * (T k) {return P<T>(x * k, y * k);}
16    P<T> operator / (T k) {return P<T>(x / k, y / k);}
17    T operator * (P<T> o) {return x * o.x + y * o.y;}
18    T operator ^ (P<T> o) {return x * o.y - y * o.x;}
19    friend ostream& operator << (ostream& o, P<T> a) {
20        return o << "(" << a.x << ", " << a.y << ")"; }
21    bool operator == (P<T> o) {
22        return sign(x - o.x) == 0 && sign(y - o.y) == 0;}
23 };
24 using Pt = P<ll>;
25 struct Line { Pt a, b; };
26 struct Cir { Pt o; double r; };
27 ll abs2(Pt a) { return a.x * a.x + a.y * a.y; }
28 double abs(Pt a) { return sqrt(abs2(a)); }
29 int ori(Pt o, Pt a, Pt b)
30 { return sign((o - a) ^ (o - b)); }
31 bool btw(Pt a, Pt b, Pt c) // c on segment ab?
32 { return ori(a, b, c) == 0 &&
33     sign((c - a) * (c - b)) <= 0; }
34 int pos(Pt a)
35 { return sign(a.y) == 0 ? sign(a.x) < 0 : a.y < 0; }
36 bool cmp(Pt a, Pt b)
37 { return pos(a) == pos(b) ? sign(a ^ b) > 0 :
38     pos(a) < pos(b); }
39 bool same_vec(Pt a, Pt b, int d) // d = 1: check dir
40 { return sign(a ^ b) == 0 && sign(a * b) > d * 2 - 2; }
41 bool same_vec(Line a, Line b, int d)
42 { return same_vec(a.b - a.a, b.b - b.a, d); }
43 Pt perp(Pt a) { return Pt(-a.y, a.x); } // CCW 90 deg
44 Pt ref(Pt a) {return pos(a) == 1 ? Pt(-a.x, -a.y) : a;}
45 // double part
46 double theta(Pt a)
47 { return normalize(atan2(a.y, a.x)); }
48 Pt unit(Pt o) { return o / abs(o); }
49 Pt rot(Pt a, double o) // CCW
50 { double c = cos(o), s = sin(o);
51     return Pt(c * a.x - s * a.y, s * a.x + c * a.y); }
52 Pt proj_vec(Pt a, Pt b, Pt c) // vector ac proj to ab
53 {return (b - a) * ((c - a) * (b - a)) / (abs2(b - a));}
54 Pt proj_pt(Pt a, Pt b, Pt c) // point c proj to ab
55 { return proj_vec(a, b, c) + a; }

```

4.2 convex hull

```

1 auto convex_hull(vector<Pt> pts) {
2     sort(all(pts), [&](Pt a, Pt b)
3     { return a.x == b.x ? a.y < b.y : a.x < b.x; });
4     vector<Pt> ans = {pts[0]};
5     for (int t = 0; t < 2; ++t, reverse(all(pts))) {
6         for (int i = 1, m = sz(ans); i < sz(pts); ++i) {
7             while (sz(ans) > m && ori(ans[sz(ans) - 2],
8                 ans.back(), pts[i]) <= 0) ans.pop_back();
9             ans.pb(pts[i]);
10        }
11    }
12    if (sz(ans) > 1) ans.pop_back();
13    return ans;
14 }

```

4.3 convex hull dist

```

1 // 計算兩個凸多邊形 A、B 之間的最短距離
2 // 兩者相交或接觸就回傳 0，否則回傳兩者邊界上最近兩點的距離
3 double convex_hull_dist(vector<Pt> A, vector<Pt> B) {
4     Pt O(0, 0); for (auto &p : B) p = O - p;
5     auto C = minkowski(A, B); // assert SZ(C) > 0
6     if (point_in_convex(C, O)) return 0;
7     double ans = point_segment_dist(C.back(), C[0], O);
8     for (int i = 0; i + 1 < sz(C); ++i)
9         ans = min(ans, point_segment_dist(C[i], C[i + 1], O));
10    return ans;
11 }

```

4.4 Minimum Enclosing Circle

```

1 Cir min_enclosing(vector<Pt> p) {
2     random_shuffle(all(p));
3     double r = 0.0;
4     Pt cent = p[0];
5     for (int i = 1; i < sz(p); ++i) {
6         if (abs2(cent - p[i]) <= r) continue;
7         cent = p[i], r = 0.0;
8         for (int j = 0; j < i; ++j) {
9             if (abs2(cent - p[j]) <= r) continue;
10            cent = (p[i] + p[j]) / 2, r = abs2(p[j] - cent);
11            for (int k = 0; k < j; ++k) {
12                if (abs2(cent - p[k]) <= r) continue;
13                cent = circenter(p[i], p[j], p[k]);
14                r = abs2(p[k] - cent);
15            }
16        }
17    }
18    return {cent, sqrt(r)};
19 }

```

4.5 Minkowski

```

1 auto minkowski(vector<Pt> a, vector<Pt> b) {
2     a = convex_hull(a), b = convex_hull(b);
3     int n = sz(a), m = sz(b);
4     vector<Pt> c = {a[0] + b[0]}, s1, s2;
5     for (int i = 0; i < n; ++i)
6         s1.pb(a[(i + 1) % n] - a[i]);
7     for (int i = 0; i < m; ++i)
8         s2.pb(b[(i + 1) % m] - b[i]);
9     for (int p1 = 0, p2 = 0; p1 < n || p2 < m;)
10         if (p2 == m || (p1 < n && sign(s1[p1] ^ s2[p2]) >= 0))
11             c.pb(c.back() + s1[p1++]);
12         else
13             c.pb(c.back() + s2[p2++]);
14     return convex_hull(c);
15 }

```

4.6 Point In Convex

```

1 // strict: if true, point on edge is considered outside
2 bool point_in_convex(vector<Pt> &C, Pt p, bool strict = true) {
3     // only works when no three points are collinear
4     int a = 1, b = sz(C) - 1, r = !strict;
5     if (sz(C) == 0) return false;
6     if (sz(C) < 3) return r && btw(C[0], C.back(), p);
7     if (ori(C[0], C[a], C[b]) > 0) swap(a, b);
8     if (ori(C[0], C[a], p) >= r || ori(C[0], C[b], p) <= -r)
9         return false;

```

```

9   while (abs(a - b) > 1) {
10       int c = (a + b) / 2;
11       (ori(C[0], C[c], p) > 0 ? b : a) = c;
12   }
13   return ori(C[a], C[b], p) < r;
14 }
15 // 回傳 p 所在的邊 {i, j}
16 // p 不在邊上時回傳 {-1, -1}
17 pair<int, int> point_on_convex_edge(vector<Pt> &C, Pt p) {
18     int a = 1, b = sz(C) - 1;
19     if (ori(C[0], C[a], C[b]) > 0) swap(a, b);
20     if (ori(C[0], C[a], p) == 0 && btw(C[0], C[a], p)) return {0, a};
21     if (ori(C[0], C[b], p) == 0 && btw(C[0], C[b], p)) return {0, b};
22     if (ori(C[0], C[a], p) > 0 || ori(C[0], C[b], p) < 0) return {-1, -1};
23     while (abs(a - b) > 1) {
24         int c = (a + b) / 2;
25         (ori(C[0], C[c], p) > 0 ? b : a) = c;
26     }
27     if (ori(C[a], C[b], p) == 0) return {a, b};
28     return {-1, -1};
29 }

```

4.7 Point In Poly

```

1 // 判斷點是否在多邊形內
2 // -1: 在外面
3 // 0: 在邊上 (只有 strict = 1 時會出現)
4 // 1: 在裡面 (strict = 0 時·在邊上也回傳 1)
5 // -1: out, 0: edge, 1: in
6 int point_in_poly(vector<Pt> poly, Pt o, int strict) {
7     int cnt = 0;
8     for (int i = 0; i < sz(poly); ++i) {
9         Pt a = poly[i], b = poly[(i + 1) % sz(poly)];
10        if (btw(o, a, b)) return !strict;
11        cnt ^= ((o.y < a.y) - (o.y - b.y)) * ori(o, a, b) > 0;
12    }
13    return cnt ? 1 : -1;
14 }

```

4.8 Point In Triangle

```

1 // S: 點集 (排序、去重)
2 // dirs[i]: 所有 p > S[i] 的向量 p - S[i] · 依極角排序
3 // 這些向量都落在半平面 (x > 0 或 x == 0 && y > 0) · 所以用外積
  比較是合法的全序
4 vector<Pt> S;
5 vector<vector<Pt>> dirs;
6 int id(Pt p) { return lower_bound(all(S), p) - S.begin(); }
7 void init(vector<Pt> pts) {
8     S = pts; sort(all(S)); S.erase(unique(all(S)), S.end());
9     int n = sz(S); dirs.assign(n, {});
10    for (int i = 0; i < n; ++i) {
11        for (int j = i + 1; j < n; ++j) dirs[i].push_back(S[j] - S[i]);
12    }
13    sort(all(dirs[i]), [](Pt u, Pt v) { return (u ^ v) > 0; });
14 }
15 // q 必須在 S 裡; 在所有 p > q 中·數 (d ^ (p - q)) < 0 與 == 0
  的個數
16 pair<int, int> cnt(Pt q, Pt d) {
17     auto &v = dirs[id(q)];
18     auto cmp = [](Pt u, Pt w) { return (u ^ w) > 0; };
19     auto lo = lower_bound(all(v), d, cmp), hi = upper_bound(all(v), d, cmp);
20     return {int(lo - v.begin()), int(hi - lo)};
21 }
22 // check if this number is calculated (p 本身是不是點集裡的點)
23 bool check(Pt p) { int i = id(p); return i < sz(S) && S[i] == p; }
24 // number of points p with a < p < b such that ori(p, a, b) < 0
25 int under(Pt a, Pt b) {
26     Pt d = b - a;
27     return cnt(a, d).first - cnt(b, d).first;
28 }
29 // number of points with a < p < b and ori(p, a, b) = 0
30 int edge(Pt a, Pt b) {
31     Pt d = b - a;
32     return cnt(a, d).second - cnt(b, d).second - check(b);
33 }
34 // number of points that strictly inside the triangle
35 int in_tri(array<Pt, 3> arr) {
36     sort(all(arr), [&](Pt i, Pt j) {
37         return i.x == j.x ? i.y < j.y : i.x < j.x; });
38     auto [a, b, c] = arr;
39     int x = ori(b, a, c);
40     if (x == 0) return 0;

```

```

41     if (x == 1) return under(a, b) + under(b, c) - under(a, c) -
        edge(a, c);
42     return under(a, c) - under(a, b) - under(b, c) - edge(a, b) -
        edge(b, c) - check(b);
43 }
44 int main() {
45     int n; cin >> n;
46     vector<Pt> pts(n);
47     for (auto &p : pts) cin >> p.x >> p.y;
48     init(pts);
49     int q; cin >> q;
50     while (q--) {
51         int i, j, k; cin >> i >> j >> k; // 三個頂點在輸入中的編號
        (0-indexed)
52         cout << in_tri({pts[i], pts[j], pts[k]}) << '\n';
53     }
54 }

```

4.9 Point Segment Dist

```

1 double point_segment_dist(Pt q0, Pt q1, Pt p) {
2     if (sign(abs(q0 - q1)) == 0) return abs(q0 - p);
3     if (sign((q1 - q0) * (p - q0)) >= 0 && sign((q0 - q1) * (p -
        q1)) >= 0)
4         return fabs(((q1 - q0) ^ (p - q0)) / abs(q0 - q1));
5     return min(abs(p - q0), abs(p - q1));
6 }

```

5 graph

5.1 2SAT

```

1 struct TWO_SAT {
2     int n, N;
3     vector<vector<int>> G, rev_G;
4     deque<bool> used;
5     vector<int> order, comp;
6     deque<bool> assignment;
7     void init(int _n) {
8         n = _n;
9         N = _n * 2;
10        G.resize(N + 5);
11        rev_G.resize(N + 5);
12    }
13    void dfs1(int v) {
14        used[v] = true;
15        for (int u : G[v]) {
16            if (!used[u])
17                dfs1(u);
18        }
19        order.push_back(v);
20    }
21    void dfs2(int v, int cl) {
22        comp[v] = cl;
23        for (int u : rev_G[v]) {
24            if (comp[u] == -1)
25                dfs2(u, cl);
26        }
27    }
28    bool solve() {
29        order.clear();
30        used.assign(N, false);
31        for (int i = 0; i < N; ++i) {
32            if (!used[i])
33                dfs1(i);
34        }
35        comp.assign(N, -1);
36        for (int i = 0, j = 0; i < N; ++i) {
37            int v = order[N - i - 1];
38            if (comp[v] == -1)
39                dfs2(v, j++);
40        }
41        assignment.assign(n, false);
42        for (int i = 0; i < N; i += 2) {
43            if (comp[i] == comp[i + 1])
44                return false;
45            assignment[i / 2] = (comp[i] > comp[i + 1]);
46        }
47        return true;
48    }
49    void add_disjunction(int a, bool na, int b, bool nb) { // A
        or B
50        // na means whether a is negative or not
51        // nb means whether b is negative or not
52        a = 2 * a ^ na;
53        b = 2 * b ^ nb;
54        int neg_a = a ^ 1;
55        int neg_b = b ^ 1;

```

```

56     G[neg_a].push_back(b);
57     G[neg_b].push_back(a);
58     rev_G[b].push_back(neg_a);
59     rev_G[a].push_back(neg_b);
60     return;
61 }
62 void get_result(vector<int>& res) {
63     res.clear();
64     for (int i = 0; i < n; i++)
65         res.push_back(assignment[i]);
66 }
67 };

```

5.2 bridge

```

1  const int N = 200005;
2  vector<int> G[N];
3  int low[N], depth[N];
4  bool vis[N];
5  vector<pair<int, int>> bridge;
6  void dfs(int v, int p) {
7      vis[v] = true;
8      low[v] = depth[v] = (p == -1 ? 1 : depth[p] + 1);
9      for (int u : G[v]) {
10         if (u == p) continue;
11         if (!vis[u]) {
12             // (v, u) 是樹邊
13             dfs(u, v);
14             low[v] = min(low[v], low[u]);
15         } else {
16             // (v, u) 是回邊
17             low[v] = min(low[v], depth[u]);
18         }
19     }
20     // v 在不依靠父邊的情況下永遠沒辦法走到它的祖先
21     // (root, -1) 不算
22     if (low[v] == depth[v] && p != -1)
23         bridge.push_back({v, p});
24 }

```

5.3 bridge bcc

```

1  const int N = 200005;
2  vector<int> G[N];
3  int low[N], depth[N];
4  bool vis[N];
5  vector<vector<int>> bcc;
6  stack<int> stk;
7  void dfs(int v, int p) {
8      stk.push(v);
9      vis[v] = true;
10     low[v] = depth[v] = (p == -1 ? 1 : depth[p] + 1);
11     for (int u : G[v]) {
12         if (u == p) continue;
13         if (!vis[u]) {
14             // (v, u) 是樹邊
15             dfs(u, v);
16             low[v] = min(low[v], low[u]);
17         } else {
18             // (v, u) 是回邊
19             low[v] = min(low[v], depth[u]);
20         }
21     }
22     // v 在不依靠父邊的情況下永遠沒辦法走到它的祖先
23     if (low[v] == depth[v]) {
24         bcc.emplace_back();
25         while (stk.top() != v) {
26             bcc.back().push_back(stk.top());
27             stk.pop();
28         }
29         bcc.back().push_back(stk.top());
30         stk.pop();
31     }
32 }

```

5.4 cut

```

1  const int N = 200005;
2  vector<int> G[N];
3  int low[N], depth[N];
4  bool vis[N], cut[N];
5  void dfs(int v, int p) {
6      vis[v] = true;
7      int child = 0;
8      low[v] = depth[v] = (p == -1 ? 1 : depth[p] + 1);
9      for (int u : G[v]) {

```

```

10         if (u == p) continue;
11         if (!vis[u]) {
12             // (v, u) 是樹邊
13             dfs(u, v);
14             child++;
15             low[v] = min(low[v], low[u]);
16             // u 無法在不經過父邊的情況下走到 v 的祖先
17             if (low[u] >= depth[v] && p != -1)
18                 cut[v] = true;
19         } else {
20             // (v, u) 是回邊
21             low[v] = min(low[v], depth[u]);
22         }
23     }
24     // 根節點有超過 2 個子節點
25     if (p == -1 && child >= 2)
26         cut[v] = true;
27 }

```

5.5 cut bcc

```

1  const int N = 200005;
2  vector<int> G[N];
3  int low[N], depth[N];
4  bool vis[N];
5  vector<vector<int>> bcc;
6  stack<int> stk;
7  void dfs(int v, int p) {
8      stk.push(v);
9      vis[v] = true;
10     low[v] = depth[v] = (p == -1 ? 1 : depth[p] + 1);
11     for (int u : G[v]) {
12         if (u == p) continue;
13         if (!vis[u]) {
14             // (v, u) 是樹邊
15             dfs(u, v);
16             low[v] = min(low[v], low[u]);
17             // u 無法在不經過父邊的情況下走到 v 的祖先
18             if (low[u] >= depth[v]) {
19                 bcc.emplace_back();
20                 while (stk.top() != u) {
21                     bcc.back().push_back(stk.top());
22                     stk.pop();
23                 }
24                 bcc.back().push_back(stk.top());
25                 stk.pop();
26                 bcc.back().push_back(v);
27             }
28         } else {
29             // (v, u) 是回邊
30             low[v] = min(low[v], depth[u]);
31         }
32     }
33 }

```

5.6 prim

```

1  const int N = 2005;
2  int dis[N][N];
3  int distfromtree[N];
4  for (int i = 1; i <= n; i++)
5      distfromtree[i] = dis[1][i];
6  int res = 0;
7  for (int i = 2; i <= n; i++) {
8      int mindist = INT_MAX, next = -1;
9      for (int j = 1; j <= n; j++) {
10         if (distfromtree[j] != 0 && distfromtree[j] < mindist) {
11             mindist = distfromtree[j];
12             next = j;
13         }
14     }
15     res += mindist;
16     for (int j = 1; j <= n; j++)
17         distfromtree[j] = min(distfromtree[j], dis[next][j]);
18 }

```

5.7 spfa

```

1  vector<int> G[N];
2  int spfa(int n, int s, int t) {
3      vector<int> dis(n, INT_MAX), cnt(n);
4      vector<bool> inq(n);
5      queue<int> q;
6      q.push(s);
7      dis[s] = 0;
8      inq[s] = true;

```

```

9   while (q.size()) {
10       int u = q.front();
11       q.pop();
12       cnt[u]++;
13       if (cnt[u] == n) // negative cycle
14           return INT_MAX;
15       inq[u] = false;
16       for (auto [v, w] : G[u]) {
17           if (dis[u] + w < dis[v]) {
18               dis[v] = dis[u] + w;
19               if (!inq[v]) {
20                   inq[v] = true;
21                   q.push(v);
22               }
23           }
24       }
25   }
26   return dis[t];
27 }

```

5.8 Strongly Connected Component

```

1   const int N = 200005;
2   vector<int> G[N], rev_G[N], order;
3   bool vis[N]; int scc[N];
4   void rev_dfs(int v) {
5       vis[v] = true;
6       for (int u : rev_G[v])
7           if (!vis[u])
8               rev_dfs(u);
9       order.push_back(v);
10  }
11  void dfs(int v, int block) {
12      scc[v] = block;
13      for (int u : G[v])
14          if (scc[u] == -1)
15              dfs(u, block);
16  }
17  void kosaraju(int n) {
18      for (int i = 1; i <= n; i++)
19          if (!vis[i])
20              rev_dfs(i);
21      int scc_cnt = 0;
22      fill(scc, scc + N, -1);
23      for (int i = n - 1; i >= 0; i--)
24          if (scc[order[i]] == -1)
25              dfs(order[i], ++scc_cnt);
26  }

```

6 math

6.1 big number

```

1   void init(string &a, string &b) {
2       while (a.size() < b.size()) a = '0' + a;
3       while (a.size() > b.size()) b = '0' + b;
4   }
5   int Compare(string &a, string &b) {
6       if (a < b) {
7           swap(a, b); return -1;
8       }
9       return 1;
10  }
11  bool del(string &a) {
12      if (a[0] == '0') {
13          a.erase(0, 1); return true;
14      }
15      return false;
16  }
17  void del_all_zero(string &a) {
18      while (del(a));
19  }
20  string add(string a, string b) {
21      init(a, b);
22      a = '0' + a; b = '0' + b;
23      for (int i = a.size() - 1; i >= 0; i--) {
24          int p = a[i] - '0', q = b[i] - '0';
25          if (p + q >= 10) {
26              a[i - 1] = '0' + (a[i - 1] - '0' + 1);
27              a[i] = '0' + (p + q - 10);
28          } else {
29              a[i] = '0' + (p + q);
30          }
31      }
32      del(a);
33      return a;
34  }
35  string sub(string a, string b) {
36      init(a, b);

```

```

37  if (a == b) return "0";
38  int flag = Compare(a, b);
39  for (int i = a.size() - 1; i >= 0; i--) {
40      int p = a[i] - '0', q = b[i] - '0';
41      if (p < q) {
42          a[i - 1] = '0' + (a[i - 1] - '0' - 1);
43          a[i] = '0' + (p - q + 10);
44      } else {
45          a[i] = '0' + (p - q);
46      }
47  }
48  del_all_zero(a);
49  if (flag == -1) a = "-" + a;
50  return a;
51 }
52 string mul(string a, string b) {
53     string res = "0";
54     init(a, b);
55     Compare(a, b);
56     del_all_zero(b);
57     for (int i = b.size() - 1; i >= 0; i--) {
58         int x = b[i] - '0';
59         if (i != b.size() - 1) a = a + '0';
60         for (int i = 0; i < x; i++)
61             res = add(a, res);
62     }
63     return res;
64 }
65 string div(string a, string b) {
66     init(a, b);
67     if (a < b) return "0";
68     del_all_zero(b);
69     string res = "0", restmp = "1", tmp = b;
70     for (int i = 0; i < (a.size() - b.size()); i++) {
71         restmp += '0';
72         tmp += '0';
73     }
74     init(a, b);
75     while (a >= b) {
76         init(a, tmp);
77         if (a >= tmp) {
78             a = sub(a, tmp);
79             res = add(res, restmp);
80         } else {
81             restmp.erase(restmp.size() - 1);
82             tmp.erase(tmp.size() - 1);
83         }
84         init(a, b);
85     }
86     return res;
87 }

```

6.2 burnside lemma

$$\sum_{k=1}^n \frac{c(k)}{n}$$

- n : 有多少種置換方式 (例如 : 旋轉方式)
- $c(k)$: 所有可能中 · 經過 k 次旋轉後 · 仍不會和別人相同方式的數量

6.3 catalan

$$C_n = \frac{1}{n+1} \binom{2n}{n}$$

6.4 Chinese Remainder Theorem coprime

```

1   vector<int> a, m;
2   int extgcd(int a, int b, int& x, int& y) {
3       if (b == 0) {
4           x = 1, y = 0;
5           return a;
6       }
7       int ret = extgcd(b, a % b, y, x);
8       y -= a / b * x;
9       return ret;
10  }
11  // n = 有幾個式子 · 求解  $x \equiv a_i \pmod{m_i}$ 
12  int CRT(int n, vector<int>& a, vector<int>& m) {
13      int p = 1, ans = 0;
14      vector<int> M(n), inv_M(n);
15      for (int i = 0; i < n; i++)
16          p *= m[i];
17      for (int i = 0; i < n; i++) {
18          M[i] = p / m[i];
19          extgcd(M[i], m[i], inv_M[i], tmp);
20          ans += a[i] * inv_M[i] * M[i];
21          ans %= p;
22      }

```

```

23 |     return (ans % p + p) % p;
24 | }

```

6.5 Chinese Remainder Theorem coprime

$n = \text{有幾個式子} \cdot \text{求解 } x \equiv a_i \pmod{m_i}$

6.6 Chinese Remainder Theorem not coprime

```

1 | int extgcd(int a, int b, int& x, int& y) {
2 |     if (b == 0) {
3 |         x = 1, y = 0;
4 |         return a;
5 |     }
6 |     int ret = extgcd(b, a % b, y, x);
7 |     y -= a / b * x;
8 |     return ret;
9 | }
10 | // 對於方程組的式子兩兩求解 · return {是否有解, {a, m}}
11 | pair<bool, pair<int, int>> CRT(int a1, int m1, int a2, int m2)
12 | {
13 |     int g = __gcd(m1, m2);
14 |     if ((a2 - a1) % g != 0)
15 |         return {0, {-1, -1}};
16 |     int x, y;
17 |     extgcd(m1, m2, x, y);
18 |     x = (a2 - a1) * x / g; // 兩者不能相反
19 |     a1 = x * m1 + a1;
20 |     m1 = m1 * m2 / g;
21 |     a1 = (a1 % m1 + m1) % m1;
22 |     return {1, {a1, m1}};
23 | }

```

6.7 count prime

```

1 | // Count the number of primes not greater than N (N <= 10^11)
2 | int primeCount(const int n) {
3 |     if (n <= 1) { return 0; }
4 |     if (n == 2) { return 1; }
5 |     const int v = sqrtl(n);
6 |     int s = (v + 1) / 2;
7 |     vector<int> smalls(s), roughs(s), skip(v + 1), larges(s);
8 |     iota(smalls.begin(), smalls.end(), 0);
9 |     for (int i = 0; i < s; i++) {
10 |         roughs[i] = 2 * i + 1;
11 |         larges[i] = (n / roughs[i] - 1) / 2;
12 |     }
13 |     const auto half = [](int n) -> int { return (n - 1) >> 1; };
14 |     int pc = 0;
15 |     for (int p = 3; p <= v; p += 2) {
16 |         if (skip[p]) { continue; }
17 |         int q = p * p;
18 |         if (1LL * q * q > n) { break; }
19 |         skip[p] = true;
20 |         for (int i = q; i <= v; i += 2 * p) skip[i] = true;
21 |         int ns = 0;
22 |         for (int k = 0; k < s; k++) {
23 |             int i = roughs[k];
24 |             if (skip[i]) { continue; }
25 |             int d = 1LL * i * p;
26 |             larges[ns] = larges[k] - (d <= v ? larges[smalls[d / 2] - pc] : smalls[half(n / d)]) + pc;
27 |             roughs[ns++] = i;
28 |         }
29 |         s = ns;
30 |         for (int i = half(v), j = v / p - 1 | 1; j >= p; j -= 2) {
31 |             int c = smalls[j / 2] - pc;
32 |             for (int e = j * p / 2; i >= e; i--) { smalls[i] -= c; }
33 |         }
34 |         pc++;
35 |     }
36 |     larges[0] += 1LL * (s + 2 * (pc - 1)) * (s - 1) / 2;
37 |     for (int k = 1; k < s; k++) { larges[0] -= larges[k]; }
38 |     for (int l = 1; l < s; l++) {
39 |         int q = roughs[l];
40 |         int M = n / q;
41 |         int e = smalls[half(M / q)] - pc;
42 |         if (e <= 1) { break; }
43 |         int t = 0;
44 |         for (int k = 1 + 1; k <= e; k++) { t += smalls[half(M / roughs[k])]; }
45 |         larges[0] += t - 1LL * (e - 1) * (pc + 1 - 1);
46 |     }
47 |     return larges[0] + 1;
48 | }

```

6.8 det

```

1 | ll Det(vector<vector<ll>> a) {
2 |     int n = a.size();
3 |     ll det = 1;
4 |     for (int i = 0; i < n; ++i) {
5 |         if (!a[i][i]) {
6 |             det = -det;
7 |             if (det < 0) det += mod;
8 |             for (int j = i + 1; j < n; ++j) if (a[j][i]) {
9 |                 swap(a[j], a[i]);
10 |                 break;
11 |             }
12 |             if (!a[i][i]) return 0;
13 |         }
14 |         det = det * a[i][i] % mod;
15 |         ll mul = mpow(a[i][i], mod - 2);
16 |         for (int j = 0; j < n; ++j)
17 |             a[i][j] = a[i][j] * mul % mod;
18 |         for (int j = 0; j < n; ++j) if (i ^ j) {
19 |             ll mul = a[j][i];
20 |             for (int k = 0; k < n; ++k) {
21 |                 a[j][k] -= a[i][k] * mul % mod;
22 |                 if (a[j][k] < 0) a[j][k] += mod;
23 |             }
24 |         }
25 |     }
26 |     return det;
27 | }

```

6.9 extgcd

```

1 | // a * p.first + b * p.second = gcd(a, b)
2 | pair<ll, ll> extgcd(ll a, ll b) {
3 |     pair<ll, ll> res, tmp;
4 |     ll f = 1, g = 1;
5 |     if (a < 0) a *= -1, f *= -1;
6 |     if (b < 0) b *= -1, g *= -1;
7 |     if (b == 0) return {f, 0};
8 |     tmp = extgcd(b, a % b);
9 |     res.first = tmp.second * f;
10 |     res.second = (tmp.first - tmp.second * (a / b)) * g;
11 |     return res;
12 | }

```

6.10 factorize

```

1 | using i64 = long long;
2 | using i128 = __int128_t;
3 | struct Factorize { // 質因數分解 N <= 10^18
4 |     i64 fmul(i64 a, i64 b, i64 p) {
5 |         return (i128)a * b % p;
6 |     }
7 |     i64 fpow(i64 a, i64 b, i64 p) {
8 |         i64 res = 1;
9 |         for (; b; b >>= 1, a = fmul(a, a, p))
10 |             if (b & 1) res = fmul(res, a, p);
11 |         return res;
12 |     }
13 |     bool check(i64 a, i64 u, i64 n, int t) {
14 |         a = fpow(a, u, n);
15 |         if (a == 0 || a == 1 || a == n - 1)
16 |             return true;
17 |         for (int i = 0; i < t; i++) {
18 |             a = fmul(a, a, n);
19 |             if (a == 1) return false;
20 |             if (a == n - 1) return true;
21 |         }
22 |         return false;
23 |     };
24 |     bool isPrime(i64 n) {
25 |         constexpr array<i64, 7> magic{2, 235, 9375, 28178, 450775,
26 |             9780504, 1795265022};
27 |         // for int: {2, 7, 61}
28 |         if (n < 2) return false;
29 |         if (n % 2 == 0) return n == 2;
30 |         i64 u = n - 1;
31 |         int t = 0;
32 |         while (u % 2 == 0) u >>= 1, t++;
33 |         for (auto v : magic) if (!check(v, u, n, t)) return false;
34 |         return true;
35 |     };
36 |     i64 PollardRho(i64 n) { // return non-trivial factor of n
37 |         if (n % 2 == 0) return 2;
38 |         i64 x = 2, y = 2, d = 1, p = 1;
39 |         auto f = [](i64 x, i64 n, i64 p) -> i64 {
40 |             return ((i128)x * x % n + p) % n;
41 |         };
42 |         while (true) {

```

```

42     x = f(x, n, p);
43     y = f(f(y, n, p), n, p);
44     d = gcd(abs(x - y), n);
45     if (d != n && d != 1) return d;
46     if (d == n) ++p;
47 }
48 }
49 i64 primeFactor(i64 n) {
50     return isPrime(n) ? n : primeFactor(PollardRho(n));
51 }
52 } ftr;
53 void solve() {
54     i64 n;
55     cin >> n;
56     vector<i64> ans;
57     while (n > 1) {
58         i64 p = ftr.primeFactor(n);
59         do {
60             ans.push_back(p);
61             n /= p;
62         } while (n % p == 0);
63     }
64     sort(ans.begin(), ans.end());
65     cout << ans.size() << ' ';
66     for (int i : ans) cout << i << ' ';
67     cout << '\n';
68 }

```

6.11 fraction

```

1 struct frac {
2     int a, b;
3     frac(int _a = 0, int _b = 1) : a(_a), b(_b) {
4         int g = __gcd(a, b);
5         a /= g; b /= g;
6         if (b < 0) { a *= -1; b *= -1; }
7     }
8 };
9 frac operator+(frac x, frac y) { return frac(x.a * y.b + y.a *
10 x.b, x.b * y.b); }
11 frac operator-(frac x, frac y) { return frac(x.a * y.b - y.a *
12 x.b, x.b * y.b); }
13 frac operator*(frac x, frac y) { return frac(x.a * y.a, x.b * y
14 .b); }
15 frac operator/(frac x, frac y) { return frac(x.a * y.b, x.b * y
16 .a); }
17 bool operator>(frac x, frac y) { return x.a * y.b > y.a * x.b;
18 }
19 bool operator<(frac x, frac y) { return x.a * y.b < y.a * x.b;
20 }
21 bool operator>=(frac x, frac y) { return x.a * y.b >= y.a * x.b
22 ; }
23 bool operator<=(frac x, frac y) { return x.a * y.b <= y.a * x.b
24 ; }
25 bool operator==(frac x, frac y) { return x.a * y.b == y.a * x.b
26 ; }
27 frac abs(frac x) { return frac(abs(x.a), abs(x.b)); }
28 ostream& operator<<(ostream &os, const frac& x) { os << x.a <<
29 "/" << x.b; return os; }

```

6.12 gaussian elimination

```

1 #define int long long
2 const int N = 25;
3 frac co[N][N];
4 // SIZE = n * (n + 1)
5 // a_1 x_1 + a_2 x_2 + ... + a_n x_n = a_{n+1}
6 void gaussian(int n) {
7     for (int i = 1; i <= n; i++) {
8         if (co[i][i].a == 0 && co[i][i].b == 1) { // swap
9             int swapid = -1;
10             for (int id = i + 1; id <= n; id++) {
11                 if (! (co[id][i].a == 0 && co[id][i].b == 1)) {
12                     swapid = id; break;
13                 }
14             }
15             if (swapid == -1)
16                 continue;
17             else {
18                 for (int j = 1; j <= n + 1; j++)
19                     swap(co[i][j], co[swapid][j]);
20             }
21         }
22         for (int id = i + 1; id <= n; id++) {
23             if (! (co[id][i].a == 0 && co[id][i].b == 1)) {
24                 frac mul = co[id][i] / co[i][i];
25                 for (int j = 1; j <= n + 1; j++)
26                     co[id][j] = co[id][j] - co[i][j] * mul;
27             }
28         }
29     }

```

```

30
31 for (int i = n; i >= 1; i--) {
32     if (! (co[i][i].a == 0 && co[i][i].b == 1)) {
33         co[i][n + 1] = co[i][n + 1] / co[i][i];
34         co[i][i] = {1, 1};
35     } else
36         continue;
37     for (int id = 1; id < i; id++) {
38         if (! (co[id][i].a == 0 && co[id][i].b == 1)) {
39             frac mul = co[id][i] / co[i][i];
40             for (int j = 1; j <= n + 1; j++)
41                 co[id][j] = co[id][j] - co[i][j] * mul;
42         }
43     }
44 }
45 bool check_inf = false, check_no = false;
46 for (int i = 1; i <= n; i++) {
47     bool allzro = true;
48     for (int j = 1; j <= n; j++)
49         if (! (co[i][j].a == 0 && co[i][j].b == 1))
50             allzro = false;
51     if (allzro) {
52         if (! (co[i][n + 1].a == 0 && co[i][n + 1].b == 1))
53             check_no = true;
54         else
55             check_inf = true;
56     }
57 }
58 if (check_no) {
59     cout << "no solutions\n";
60 } else if (check_inf) {
61     cout << "infinite many solutions\n";
62 } else {
63     for (int i = 1; i <= n; i++)
64         cout << "X" << i << " = " << co[i][n + 1] << '\n';
65 }
66 }

```

6.13 josephus

```

1 // 有 n 個人，第偶數個報數的人被刪掉，問第 k 個被踢掉的是誰
2 int solve(int n, int k) {
3     if (n == 1) return 1;
4     if (k <= (n + 1) / 2) {
5         if (2 * k > n)
6             return 2 * k % n;
7         else
8             return 2 * k;
9     } else {
10         int res = solve(n / 2, k - (n + 1) / 2);
11         if (n & 1)
12             return 2 * res + 1;
13         else
14             return 2 * res - 1;
15     }
16 }

```

6.14 lucas

```

1 // 對於很大的 C^n_m 對質數 p 取模，只要 p 不大就可以用
2 int Lucas(int n, int m, int p) {
3     if (m == 0) return 1;
4     return (C(n % p, m % p, p) * Lucas(n / p, m / p, p) % p);
5 }

```

6.15 math block

```

1 // the sum of floor(n / i) for i = 1 ~ n
2 int f(int n) {
3     int ans = 0;
4     int l = 1, r = 0;
5     while (l <= n) {
6         r = n / (n / l);
7         ans += (r - l + 1) * (n / l);
8         l = r + 1;
9     }
10    return ans;
11 }
12 // the sum of ceil(n / i) for i = 1 ~ n
13 int f(int n) {
14     int ans = 0;
15     int l = 1, r = 0;
16     while (l < n) {
17         r = (n - 1) / ((n - 1) / l);
18         ans += (r - l + 1) * ((n + 1 - 1) / l);
19         l = r + 1;
20     }

```

```

21 | if (l == n) ans += (n + l - 1) / l;
22 | return ans;
23 | }

```

6.16 miller rabin

```

1 | struct Miller_Rabin {
2 |     static constexpr uint32_t ws32[3] = {2, 7, 61};
3 |     static constexpr uint64_t ws64[7] = {2, 325, 9375, 28178,
4 |         450775, 9780504, 1795265022};
5 |     template <class uint>
6 |     static constexpr uint fast_pow(uint b, uint p, uint mod) {
7 |         using Uint = conditional_t<is_same_v<uint, uint32_t>,
8 |             uint64_t, __uint128_t>;
9 |         uint ret{1};
10 |         for (; p; p >>= 1) {
11 |             if (p & 1) ret = Uint{ret} * b % mod;
12 |             b = Uint{b} * b % mod;
13 |         }
14 |         return ret;
15 |     }
16 |     template <class uint, class = enable_if_t<is_unsigned_v<uint>
17 |         >>>
18 |     static constexpr bool is_prime(uint n) {
19 |         const auto& witness = []() -> const auto& {
20 |             if constexpr (is_same_v<uint, uint32_t>) return ws32;
21 |             else return ws64;
22 |         }();
23 |         using Uint = conditional_t<is_same_v<uint, uint32_t>,
24 |             uint64_t, __uint128_t>;
25 |         if (n < 3 || !(n & 1)) return n == 2;
26 |         auto u = n - 1, t = (uint) __builtin_ctzll(u);
27 |         u >>= t;
28 |         for (auto x : witness) {
29 |             auto a = x % n;
30 |             if (a == 0 || a == 1 || a == n - 1) continue;
31 |             auto v = fast_pow(a, u, n);
32 |             uint i = 0;
33 |             for (; v != 1 && v != n - 1 && i < t; i++) v = Uint{v} *
34 |                 v % n;
35 |             if (v != n - 1 && i > 0) return false;
36 |         }
37 |         return true;
38 |     }
39 |     template <class I, class = enable_if_t<is_integral_v<I>>>
40 |     constexpr bool operator()(I x) const {
41 |         if constexpr (is_signed_v<I>) if (x < 0) return false;
42 |         using U = make_unsigned_t<I>;
43 |         return is_prime((U)x);
44 |     }
45 | } is_prime;

```

6.17 phi

```

1 | // phi(n) = 小於 n 並與 n 互質的正整數數量。
2 | // O(sqrt(n)) · 回傳 phi(n)
3 | int phi(int n) {
4 |     int ret = n;
5 |     for (int i = 2; i * i <= n; i++) {
6 |         if (n % i == 0) {
7 |             while (n % i == 0) n /= i;
8 |             ret = ret * (i - 1) / i;
9 |         }
10 |     }
11 |     if (n > 1) ret = ret * (n - 1) / n;
12 |     return ret;
13 | }
14 | // O(n log n) · 回傳 1 ~ n 的 phi 值
15 | vector<int> phi_1_to_n(int n) {
16 |     vector<int> phi(n + 1);
17 |     phi[0] = 0; phi[1] = 1;
18 |     for (int i = 2; i <= n; i++) phi[i] = i - 1;
19 |     for (int i = 2; i <= n; i++)
20 |         for (int j = 2 * i; j <= n; j += i) // 枚舉所有倍數
21 |             phi[j] -= phi[i];
22 |     return phi;
23 | }

```

6.18 phi lemma

若 a, m 互質，則：

$$a^n \bmod m = a^{n \bmod \varphi(m)} \bmod m$$

若 a, m 可能是任何數，則：

$$a^{\varphi(m) + [n \bmod \varphi(m)]} \bmod m$$

6.19 primes er

```

1 | // Time Complexity : O(nloglogn)
2 | const int N = 200005;
3 | bool is_prime[N];
4 | vector<int> primes;
5 | void build() {
6 |     fill(is_prime, is_prime + N, true);
7 |     is_prime[0] = is_prime[1] = false;
8 |     for (int i = 2; i <= 200000; i++) {
9 |         if (is_prime[i]) {
10 |             primes.push_back(i);
11 |             if (1ll * i * i <= 200000)
12 |                 for (int j = i * i; j <= 200000; j += i)
13 |                     is_prime[j] = false;
14 |         }
15 |     }
16 | }

```

6.20 primes linear

```

1 | // Time Complexity : O(n)
2 | // if Lpf(i) = i, then it means that i is a prime.
3 | // else Lpf(i) is the smallest prime factor.
4 | // The 199999-th of the prime is 2750131
5 | bool is_prime[2750135];
6 | int lpf[2750135];
7 | vector<int> primes;
8 | void init() {
9 |     fill(is_prime, is_prime + 2750135, true);
10 |     for (int i = 2; i <= 2750131; i++) {
11 |         if (is_prime[i]) {
12 |             primes.push_back(i);
13 |             lpf[i] = i;
14 |         }
15 |         for (int p : primes) {
16 |             if (p * i > 2750131) break;
17 |             is_prime[p * i] = false;
18 |             lpf[p * i] = p;
19 |             if (i % p == 0) break;
20 |         }
21 |     }
22 | }

```

6.21 two equations

$$\begin{cases} ax + by = c \\ dx + ey = f \end{cases} = \begin{cases} x = \frac{ed-bf}{ad-bc} \\ y = \frac{af-cd}{ad-bc} \end{cases}$$

若 $x = \frac{a}{0}$ 且 $y = \frac{0}{0}$ ，則代表無限多組解。若 $x = \frac{a}{0}$ 且 $y = \frac{b}{0}$ ，則代表無解。

7 other

7.1 aliens dp

```

1 | const int N = 2000005;
2 | int n, k;
3 | int a[N];
4 | pii dp0[N], dp1[N];
5 | pii calc(int c) {
6 |     dp0[1] = {0, 0}; dp1[1] = {c - a[1], 1};
7 |     for (int i = 2; i <= n; i++) {
8 |         dp0[i] = max(dp0[i - 1], dp1[i - 1] + make_pair(+a[i], 0));
9 |         dp1[i] = max(dp1[i - 1], dp0[i - 1] + make_pair(-a[i] + c,
10 |             1));
11 |     }
12 |     return dp0[n];
13 | }
14 | void solve() {
15 |     pii p = calc(0);
16 |     if (p.S <= k) {
17 |         cout << p.F << "\n";
18 |         return;
19 |     }
20 |     int l = -1e5, r = 1e6; pii res;
21 |     while (l + 1 < r) {
22 |         int mid = (l + r) >> 1;
23 |         res = calc(mid);
24 |         if (res.S == k) {
25 |             cout << res.F - mid * k << "\n";
26 |             return;
27 |         } else if (res.S > k)
28 |             r = mid;
29 |         else
30 |             l = mid;

```

```

30 }
31 pii p1 = calc(l), p2 = calc(r);
32 int dp_1 = p1.F - 1 * p1.S;
33 int dp_2 = p2.F - r * p2.S;
34 cout << (dp_1 + (dp_2 - dp_1) * (k - p1.S) / (p2.S - p1.S))
    << "\n";
35 }

```

7.2 all LCS

```

1 void all_lcs(string s, string t) { // 0-base
2     vector<int> h(sz(t));
3     iota(all(h), 0);
4     for (int a = 0; a < sz(s); ++a) {
5         int v = -1;
6         for (int c = 0; c < sz(t); ++c)
7             if (s[a] == t[c] || h[c] < v)
8                 swap(h[c], v);
9         // LCS(s[0, a], t[b, c]) =
10        // c - b + 1 - sum([h[i] >= b] | i <= c)
11        // h[i] might become -1 !!
12    }
13 }

```

7.3 dc dp

```

1 const int N = 3005;
2 int n, k, a[N], pre[N], dp[N][N];
3 int cost(int l, int r) {}
4 void dc(int j, int l, int r, int best_l, int best_r) {
5     if (l > r) return;
6     int mid = (l + r) >> 1, pos = 1;
7     for (int i = best_l; i <= best_r; i++) {
8         if (dp[i][j - 1] + cost(i + 1, mid) < dp[mid][j]) {
9             pos = i;
10            dp[mid][j] = dp[i][j - 1] + cost(i + 1, mid);
11        }
12    }
13    dc(j, l, mid - 1, best_l, pos);
14    dc(j, mid + 1, r, pos, best_r);
15 }
16 void solve() {
17     for (int i = 1; i <= n; i++)
18         dp[i][1] = pre[i] * pre[i];
19     for (int j = 2; j <= k; j++)
20         dc(j, j, n, j - 1, n);
21 }

```

7.4 digit dp

```

1 // 原題：計算 [L, r] 中兩兩相鄰數字不同的數字數量
2 // 注意邊界
3 int dp[20][10]; // i 位數 · 尾數是 j
4 void init() {
5     for (int j = 0; j <= 9; j++) dp[1][j] = 1;
6     for (int i = 2; i <= 18; i++)
7         for (int now = 0; now <= 9; now++) // 這個位數的數字
8             for (int pre = 0; pre <= 9; pre++) // 上個位數的數字
9                 if (now != pre)
10                    dp[i][now] += dp[i - 1][pre];
11 }
12 int calc(int x) { // 小於等於 x 的數量
13     if (x == 0) return 0;
14     int arr[20] = {}, len = 0;
15     while (x > 0) arr[++len] = x % 10, x /= 10;
16     // 計算 len 位數 的數量
17     int cnt = 0, pre = -1;
18     for (int idx = len; idx >= 1; idx--) {
19         for (int now = (idx == len); now < arr[idx]; now++)
20             if (now != pre)
21                 cnt += dp[idx][now];
22         if (arr[idx] == pre) break;
23         pre = arr[idx];
24         if (idx == 1) cnt++; // 特判
25     }
26     // 計算 小於 len 位數 的數量
27     for (int i = 1; i < len; i++)
28         for (int j = 1; j <= 9; j++)
29             cnt += dp[i][j];
30     return cnt;
31 }

```

7.5 geometry

```

1 double eps = 0.000001;
2 pdd operator + (pdd a, pdd b) { return {a.X + b.X, a.Y + b.Y}; }
3 pdd operator - (pdd a, pdd b) { return {a.X - b.X, a.Y - b.Y}; }
4 double dot(pdd a, pdd b) { return a.X * b.X + a.Y * b.Y; }
5 double cross(pdd a, pdd b) { return a.X * b.Y - a.Y * b.X; }
6 int sign(double x) { return (fabs(x) < eps ? 0 : (x > 0 ? 1 : -1)); }
7 int ori(pdd a, pdd b, pdd c) { return sign(cross(b - a, c - a)); }
8 // c is between a and b
9 bool btw(pdd a, pdd b, pdd c) {
10     return (sign(cross(a - c, b - c)) == 0 && sign(dot(a - c, b - c)) <= 0);
11 }
12 // ab is touch cd
13 bool banana(pdd a, pdd b, pdd c, pdd d) {
14     int a123 = ori(a, b, c), a124 = ori(a, b, d);
15     int a341 = ori(c, d, a), a342 = ori(c, d, b);
16     if (btw(a, b, c) || btw(a, b, d) || btw(c, d, a) || btw(c, d, b))
17         return true;
18     return (a123 * a124 < 0 && a341 * a342 < 0);
19 }
20 vector<pdd> convex_hull(vector<pdd> pts) {
21     sort(pts.begin(), pts.end());
22     pts.resize(unique(pts.begin(), pts.end()) - pts.begin());
23     vector<pdd> hull(1, pts[0]);
24     for (int t = 0; t < 2; t++) {
25         int sz = hull.size();
26         for (int i = 1; i < (int)pts.size(); i++) {
27             while ((int)hull.size() > sz && ori(hull[(int)hull.size() - 2], hull.back(), pts[i]) < 0)
28                 hull.pop_back();
29             hull.push_back(pts[i]);
30         }
31         reverse(pts.begin(), pts.end());
32     }
33     hull.pop_back();
34     return hull;
35 }
36 double area(vector<pdd> pt) {
37     double cnt = 0.0;
38     for (int i = 0; i < (int)pt.size() - 1; i++)
39         cnt += cross(pt[i], pt[i + 1]);
40     cnt += cross(pt[(int)pt.size() - 1], pt[0]);
41     return fabs(cnt);
42 }
43 void solve() {
44     int n;
45     cin >> n;
46     vector<pdd> pts;
47     pts.resize(n);
48     for (int i = 0; i < n; i++)
49         cin >> pts[i].X >> pts[i].Y;
50     int ans = area(convex_hull(pts));
51     if (ans % 2 == 1)
52         cout << ans / 2 << ".5\n";
53     else
54         cout << ans / 2 << ".0\n";
55 }

```

7.6 line container

```

1 // only works for integer coordinates!! maintain max
2 struct Line {
3     mutable ll a, b, p;
4     bool operator<(const Line &rhs) const { return a < rhs.a; }
5     bool operator<(ll x) const { return p < x; }
6 };
7 struct DynamicHull : multiset<Line, less<>> {
8     static const ll kInf = 1e18;
9     ll Div(ll a, ll b) { return a / b - ((a ^ b) < 0 && a % b); }
10    bool isect(iterator x, iterator y) {
11        if (y == end()) { x->p = kInf; return 0; }
12        if (x->a == y->a) x->p = x->b > y->b ? kInf : -kInf;
13        else x->p = Div(y->b - x->b, x->a - y->a);
14        return x->p >= y->p;
15    }
16    void addline(ll a, ll b) { // ax + b
17        auto z = insert({a, b, 0}); y = z++, x = y;
18        while (isect(y, z)) z = erase(z);
19        if (x != begin() && isect(--x, y)) isect(x, y = erase(y));
20        while ((y = x) != begin() && (--x)->p >= y->p) isect(x, erase(y));
21    }
22    ll query(ll x) {
23        auto l = *lower_bound(x);
24        return l.a * x + l.b;
25    }

```

26 };

7.7 mo

```

1 struct MoSolver {
2     struct query {
3         int l, r, id;
4         bool operator < (const query &o) {
5             if (1 / C == o.l / C) return (1 / C) & 1 ? r > o.r : r <
              o.r;
6             return 1 / C < o.l / C;
7         }
8     };
9     int cur_ans;
10    vector<int> ans;
11    void add(int x) {
12        // do something
13    }
14    void sub(int x) {
15        // do something
16    }
17    vector<query> Q;
18    void add_query(int l, int r, int id) {
19        // [l, r)
20        Q.push_back({l, r, id});
21        ans.push_back(0);
22    }
23    void run() {
24        sort(Q.begin(), Q.end());
25        int pl = 0, pr = 0;
26        cur_ans = 0;
27        for (query &i : Q) {
28            while (pl > i.l)
29                add(a[--pl]);
30            while (pr < i.r)
31                add(a[pr++]);
32            while (pl < i.l)
33                sub(a[pl++]);
34            while (pr > i.r)
35                sub(a[pr--]);
36            ans[i.id] = cur;
37        }
38    }
39 };

```

7.8 number split

```

1 // dp[i][x] = 要將整數 x 拆成 i 堆的「組合數」
2 dp[i + 1][x + 1] += dp[i][x]; // 創造新的一堆
3 dp[i][x + i] += dp[i][x]; // 把每一堆都增加 1

```

7.9 python

```

1 from [decimal, fractions, math, random] import *
2 arr = list(map(int, input().split())) # input
3 setcontext(Context(prec=10, Emax=MAX_EMAX, rounding=ROUND_FLOOR
4 ))
5 Decimal('1.1') / Decimal('0.2')
6 Fraction(3, 7)
7 Fraction(Decimal('1.14'))
8 Fraction('1.2').limit_denominator(4).numerator
9 Fraction(cos(pi / 3)).limit_denominator()
10 S = set(), S.add((a, b)), S.remove((a, b)) # set
11 if not (a, b) in S:
12     D = dict(), D[(a, b)] = 1, del D[(a, b)] # dict
13 for (a, b) in D.items():
14     arr = [randint(1, C) for i in range(N)]
15     choice([8, 6, 4, 1]) # random pick one

```

7.10 sos dp

```

1 // 時間複雜度為 O(n * 2 ^ n)
2 // 計算 dp[i] = i 所有 bit mask 子集的和
3 for (int i = 0; i < n; i++)
4     for (int mask = 0; mask < (1 << n); mask++)
5         if ((mask >> i) & 1)
6             dp[mask] += dp[mask ^ (1 << i)];

```

7.11 subset

```

1 for (int i = 0; i < (1 << n); i++){
2     for (int m = i; m; m = (m - 1) & i){
3         // m 是 i 的非空子集
4     }
5 }

```

7.12 xor basis

```

1 // 給一數字集合 S · 求能 XOR 出多少數字
2 // 答案等於 2 ^ {basis 的大小}
3 vector<int> basis;
4 void add_vector(int x) {
5     for (int v : basis) x = min(x, x ^ v);
6     if (x) basis.push_back(x);
7 }
8 // 給一數字集合 S · 求能不能 XOR 出 x
9 bool check(int x) {
10    for (int v : basis) x = min(x, x ^ v);
11    return x;
12 }
13 // 給一數字集合 S · 求 XOR 出最大的數字
14 int get_max() {
15     int ans = 0;
16     for (auto v : basis) ans = max(ans, ans ^ v);
17     return ans;
18 }

```

8 polynomial

8.1 FFT

```

1 using T = complex<double>;
2 const double PI = acos(-1);
3 struct FFT {
4     T w[N];
5     FFT() {
6         for (int i = 0; i < N; ++i) w[i] = T(cos(2 * PI * i / N),
              sin(2 * PI * i / N));
7     }
8     void operator()(vector<T>& a, bool inv = false) {
9         int n = sz(a);
10        for (int i = 1, j = 0; i < n; ++i) {
11            for (int k = n >> 1; (j ^ k) < i; k >>= 1);
12            if (i < j) swap(a[i], a[j]);
13        }
14        for (int L = 1; L < n; L <= 1)
15            for (int i = 0, st = N / (2 * L); i < n; i += 2 * L)
16                for (int j = 0; j < L; ++j) {
17                    T x = a[i + j + L] * w[j * st];
18                    a[i + j + L] = a[i + j] - x;
19                    a[i + j] += x;
20                }
21        if (inv) {
22            reverse(1 + all(a));
23            T invn = 1.0 / n;
24            for (int i = 0; i < n; ++i) a[i] = a[i] * invn;
25        }
26    }
27 } ntt;
28 // after mul, round i.real()
29 // for (int i = 0; i < m; ++i) a[i] = llround(a[i].real());

```

8.2 NTT

```

1 // mul, add, sub, Pow
2 struct NTT {
3     int w[N];
4     NTT() {
5         int dw = Pow(G, (mod - 1) / N);
6         w[0] = 1;
7         for (int i = 1; i < N; ++i)
8             w[i] = mul(w[i - 1], dw);
9     } // 0 <= a[i] < P
10    void operator()(vector<int>& a, bool inv = false) {
11        int n = sz(a);
12        for (int j = 1, x = 0; j < n - 1; ++j) {
13            for (int k = n >> 1; (x ^ k) < j; k >>= 1);
14            if (j < x) swap(a[x], a[j]);
15        }
16        for (int L = 2; L <= n; L <= 1) {
17            int dx = N / L, dl = L >> 1;
18            for (int i = 0; i < n; i += L) {

```

```

19     for (int j = i, x = 0; j < i + d1; ++j, x += dx) {
20         int tmp = mul(a[j + d1], w[x]);
21         a[j + d1] = sub(a[j], tmp);
22         a[j] = add(a[j], tmp);
23     }
24 }
25 }
26 if (inv) {
27     reverse(1 + all(a));
28     int invn = Pow(n, mod - 2);
29     for (int i = 0; i < n; ++i)
30         a[i] = mul(a[i], invn);
31 }
32 }
33 } ntt;

```

8.3 NTT prime

Prime	Root	Prime	Root
7681	17	167772161	3
12289	11	104857601	3
40961	3	985661441	3
65537	3	998244353	3
786433	10	1107296257	10
5767169	3	2013265921	31
7340033	3	2810183681	11
23068673	3	2885681153	3
469762049	3	605028353	3
2061584302081	7	1945555039024054273	5
2748779069441	3	9223372036737335297	3

8.4 operation

```

1 typedef vector<int> Poly;
2 Poly Mul(Poly a, Poly b, int bound = N) {
3     int m = sz(a) + sz(b) - 1, n = 1;
4     while (n < m) n <= 1;
5     a.resize(n), b.resize(n);
6     ntt(a), ntt(b);
7     for (int i = 0; i < n; ++i) a[i] = mul(a[i], b[i]);
8     ntt(a, true), a.resize(min(m, bound));
9     return a;
10 }
11 Poly Inverse(Poly a) {
12     // O(NLogN), a[0] != 0
13     int n = sz(a);
14     Poly res(1, Pow(a[0], mod - 2));
15     for (int m = 1; m < n; m <= 1) {
16         if (n < m * 2) a.resize(m * 2);
17         Poly v1(a.begin(), a.begin() + m * 2), v2 = res;
18         v1.resize(m * 4), v2.resize(m * 4);
19         ntt(v1), ntt(v2);
20         for (int i = 0; i < m * 4; ++i)
21             v1[i] = mul(mul(v1[i], v2[i]), v2[i]);
22         ntt(v1, true);
23         res.resize(m * 2);
24         for (int i = 0; i < m; ++i)
25             res[i] = add(res[i], res[i]);
26         for (int i = 0; i < m * 2; ++i)
27             res[i] = sub(res[i], v1[i]);
28     }
29     res.resize(n);
30     return res;
31 }
32 pair<Poly, Poly> Divide(Poly a, Poly b) {
33     // a = bQ + R, O(NLogN), b.back() != 0
34     int n = sz(a), m = sz(b), k = n - m + 1;
35     if (n < m) return {{0}, a};
36     Poly ra = a, rb = b;
37     reverse(all(ra)), ra.resize(k);
38     reverse(all(rb)), rb.resize(k);
39     Poly Q = Mul(ra, Inverse(rb), k);
40     reverse(all(Q));
41     Poly res = Mul(b, Q), R(m - 1);
42     for (int i = 0; i < m - 1; ++i)
43         R[i] = sub(a[i], res[i]);
44     return {Q, R};
45 }
46 Poly SqrtImpl(Poly a) {
47     if (a.empty()) return {0};
48     int z = QuadraticResidue(a[0], mod), n = sz(a);
49     if (z == -1) return {-1};
50     Poly q(1, z);
51     const int inv2 = (mod + 1) / 2;
52     for (int m = 1; m < n; m <= 1) {
53         if (n < m * 2) a.resize(m * 2);
54         q.resize(m * 2);
55         Poly f2 = Mul(q, q, m * 2);
56         for (int i = 0; i < m * 2; ++i)
57             f2[i] = sub(f2[i], a[i]);
58         f2 = Mul(f2, Inverse(q), m * 2);
59         for (int i = 0; i < m * 2; ++i)

```

```

60         q[i] = sub(q[i], mul(f2[i], inv2));
61     }
62     q.resize(n);
63     return q;
64 }
65 Poly Sqrt(Poly a) {
66     // O(NLogN), return {-1} if not exists
67     int n = sz(a), m = 0;
68     while (m < n && a[m] == 0) m++;
69     if (m == n) return Poly(n);
70     if (m & 1) return {-1};
71     Poly s = SqrtImpl(Poly(a.begin() + m, a.end()));
72     if (s[0] == -1) return {-1};
73     Poly res(n);
74     for (int i = 0; i < sz(s); ++i)
75         res[i + m / 2] = s[i];
76     return res;
77 }
78 Poly Derivative(Poly a) {
79     int n = sz(a);
80     Poly res(n - 1);
81     for (int i = 0; i < n - 1; ++i)
82         res[i] = mul(a[i + 1], i + 1);
83     return res;
84 }
85 Poly Integral(Poly a) {
86     int n = sz(a);
87     Poly res(n + 1);
88     for (int i = 0; i < n; ++i)
89         res[i + 1] = mul(a[i], Pow(i + 1, mod - 2));
90     return res;
91 }
92 Poly Ln(Poly a) {
93     // O(NLogN), a[0] = 1
94     int n = sz(a);
95     if (n == 1) return {0};
96     Poly d = Derivative(a);
97     a.pop_back();
98     return Integral(Mul(d, Inverse(a), n - 1));
99 }
100 Poly Exp(Poly a) {
101     // O(NLogN), a[0] = 0
102     int n = sz(a);
103     Poly q(1, 1);
104     a[0] = add(a[0], 1);
105     for (int m = 1; m < n; m <= 1) {
106         if (n < m * 2) a.resize(m * 2);
107         Poly g(a.begin(), a.begin() + m * 2), h(all(q));
108         h.resize(m * 2), h = Ln(h);
109         for (int i = 0; i < m * 2; ++i)
110             g[i] = sub(g[i], h[i]);
111         q = Mul(g, q, m * 2);
112     }
113     q.resize(n);
114     return q;
115 }
116 Poly PolyPow(Poly a, ll k) {
117     int n = sz(a), m = 0;
118     Poly ans(n, 0);
119     while (m < n && a[m] == 0) m++;
120     if (k && m && (k >= n || k * m >= n)) return ans;
121     if (m == n) return ans[0] = 1, ans;
122     int lead = m * k;
123     Poly b(a.begin() + m, a.end());
124     int base = Pow(b[0], k), inv = Pow(b[0], mod - 2);
125     for (int i = 0; i < n - m; ++i)
126         b[i] = mul(b[i], inv);
127     b = Ln(b);
128     for (int i = 0; i < n - m; ++i)
129         b[i] = mul(b[i], k % mod);
130     b = Exp(b);
131     for (int i = lead; i < n; ++i)
132         ans[i] = mul(b[i - lead], base);
133     return ans;
134 }
135 vector<int> Evaluate(Poly a, vector<int> x) {
136     if (x.empty()) return {};
137     int n = sz(x);
138     vector<Poly> up(n * 2);
139     for (int i = 0; i < n; ++i)
140         up[i + n] = {sub(0, x[i]), 1};
141     for (int i = n - 1; i > 0; --i)
142         up[i] = Mul(up[i * 2], up[i * 2 + 1]);
143     vector<Poly> down(n * 2);
144     down[1] = Divide(a, up[1]).second;
145     for (int i = 2; i < n * 2; ++i)
146         down[i] = Divide(down[i >> 1], up[i]).second;
147     Poly y(n);
148     for (int i = 0; i < n; ++i) y[i] = down[i + n][0];
149     return y;
150 }
151 Poly Interpolate(vector<int> x, vector<int> y) {
152     int n = sz(x);
153     vector<Poly> up(n * 2);
154     for (int i = 0; i < n; ++i)
155         up[i + n] = {sub(0, x[i]), 1};
156     for (int i = n - 1; i > 0; --i)

```

```

157     up[i] = Mul(up[i * 2], up[i * 2 + 1]);
158 Poly a = Evaluate(Derivative(up[1]), x);
159 for (int i = 0; i < n; ++i)
160     a[i] = mul(y[i], Pow(a[i], mod - 2));
161 vector<Poly> down(n * 2);
162 for (int i = 0; i < n; ++i) down[i + n] = {a[i]};
163 for (int i = n - 1; i > 0; --i) {
164     Poly lhs = Mul(down[i * 2], up[i * 2 + 1]);
165     Poly rhs = Mul(down[i * 2 + 1], up[i * 2]);
166     down[i].resize(sz(lhs));
167     for (int j = 0; j < sz(lhs); ++j)
168         down[i][j] = add(lhs[j], rhs[j]);
169 }
170 return down[1];
171 }
Poly TaylorShift(Poly a, int c) {
172     // return sum a_i(x + c)^i;
173     // fac[i] = i!, facp[i] = inv(i!)
174     int n = sz(a);
175     for (int i = 0; i < n; ++i) a[i] = mul(a[i], fac[i]);
176     reverse(all(a));
177     Poly b(n);
178     int w = 1;
179     for (int i = 0; i < n; ++i)
180         b[i] = mul(facp[i], w, w = mul(w, c));
181     a = Mul(a, b, n), reverse(all(a));
182     for (int i = 0; i < n; ++i) a[i] = mul(a[i], facp[i]);
183     return a;
184 }
vector<int> SamplingShift(vector<int> a, int c, int m){
186     // given f(0), f(1), ..., f(n - 1)
187     // return f(c), f(c + 1), ..., f(c + m - 1)
188     int n = sz(a); // 4d649d
189     for (int i = 0; i < n; ++i) a[i] = mul(a[i], facp[i]);
190     Poly b(n);
191     for (int i = 0; i < n; ++i) {
192         b[i] = facp[i];
193         if (i & 1) b[i] = sub(0, b[i]);
194     }
195     a = Mul(a, b, n);
196     for (int i = 0; i < n; ++i) a[i] = mul(a[i], fac[i]);
197     reverse(all(a));
198     int w = 1;
199     for (int i = 0; i < n; ++i)
200         b[i] = mul(facp[i], w, w = mul(w, sub(c, i)));
201     a = Mul(a, b, n);
202     reverse(all(a));
203     for (int i = 0; i < n; ++i) a[i] = mul(a[i], facp[i]);
204     a.resize(m), b.resize(m);
205     for (int i = 0; i < m; ++i) b[i] = facp[i];
206     a = Mul(a, b, m);
207     for (int i = 0; i < m; ++i) a[i] = mul(a[i], fac[i]);
208     return a;
209 }
Poly power_proj(vector<int> w, Poly f, int m) {
210     // return sum_j w_j[x^j]f^i for i=0,1,...,m
211     int n = 1;
212     while (n < sz(f)) n <= 1;
213     f.resize(n), w.resize(n), reverse(all(w));
214     int k = 1, n2 = 2 * n, n4 = 4 * n;
215     Poly _P(n2), _Q(n2);
216     for (int i = 0; i < n; ++i)
217         _P[i] = w[i], _Q[i] = sub(0, f[i]);
218     while (n > 1) {
219         Poly R(n2);
220         for (int i = 0; i < n2; ++i)
221             R[i] = (i & 1 ? sub(0, _Q[i]) : _Q[i]);
222         Poly PQ = Mul(_P, R), QQ = Mul(_Q, R);
223         PQ.resize(n4), QQ.resize(n4);
224         for (int i = 0, j = n2; i < n2; ++i, ++j) {
225             PQ[j] = add(PQ[j], _P[i]);
226             QQ[j] = add(QQ[j], add(_Q[i], R[i]));
227         }
228         fill(all(_P), 0), fill(all(_Q), 0);
229         for (int j = 0; j < 2 * k; ++j)
230             for (int i = 0; i < n / 2; ++i) {
231                 _P[n * j + i] = PQ[2 * n * j + 2 * i + 1];
232                 _Q[n * j + i] = QQ[2 * n * j + 2 * i + 0];
233             }
234         n /= 2, k *= 2;
235     }
236     Poly p(k);
237     for (int i = 0; i < k; ++i) p[i] = _P[2 * i];
238     reverse(all(p)), p.resize(m + 1);
239     return p;
240 }

```

9 string

9.1 hash

```
const int p = 1e6 + 3;
```

```

2 const int mod = 1e9 + 7;
3 int hash(string &s) {
4     int res = 0;
5     for (int i = 0; i < s.size(); i++) {
6         res *= p;
7         res += s[i] - 'a' + 1;
8         res %= mod;
9     }
10    return res;
11 }
12 int pow_p[1000005];
13 vector<int> rh;
14 void build(string &s) {
15     rh.resize(s.size() + 1);
16     pow_p[0] = 1;
17     for (int i = 1; i <= s.size(); i++)
18         pow_p[i] = (pow_p[i - 1] * p) % mod;
19     rh[0] = s[0] - 'a' + 1;
20     for (int i = 1; i < s.size(); i++) {
21         rh[i] = rh[i - 1] * p + (s[i] - 'a' + 1);
22         rh[i] %= mod;
23     }
24 }
25 int query(int l, int r) {
26     int res = rh[r] - (l > 0 ? rh[l - 1] * pow_p[r - l + 1] : 0);
27     res = (res % mod + mod) % mod;
28     return res;
29 }

```

9.2 kmp

```

1 vector<int> F;
2 void build_failure_function(string &s) {
3     F.clear();
4     F.resize(s.size(), -1);
5     for (int i = 1; i < s.size(); i++) {
6         int j = F[i - 1];
7         while (j != -1 && s[i] != s[j + 1])
8             j = F[j];
9         if (s[i] == s[j + 1])
10            F[i] = j + 1;
11     }
12 }
13 void KMP_matching(string &a, string &b) {
14     for (int i = 0, j = -1; i < a.size(); i++) {
15         while (j != -1 && a[i] != b[j + 1])
16             j = F[j];
17         if (a[i] == b[j + 1])
18             j++;
19         if (j + 1 == b.size()) {
20             cout << "found a matching start at " << i - j << "\n";
21             j = F[j];
22         }
23     }
24 }

```

9.3 manacher

```

1 string s;
2 int extend(int l, int r, int N) {
3     int i = 0;
4     while (1 - i >= 0 && r + i < N && s[1 - i] == s[r + i])
5         i++;
6     return i;
7 }
8 int Longest_Palindromic_Substring(string &t) {
9     int N = t.length();
10    s.resize(2 * N + 1, '$');
11    for (int i = 0; i < N; i++)
12        s[2 * i + 1] = t[i];
13    N = 2 * N + 1;
14    vector<int> res;
15    res.resize(N, 0);
16    res[0] = 1;
17    for (int i = 1, mid = 0, R = 0; i < N; i++) {
18        int j = mid - (i - mid);
19        int lst = R + 1 - i;
20        if (i > R) {
21            res[i] = extend(i, i, N);
22            mid = i;
23            R = i + res[i] - 1;
24        } else if (res[j] == lst) {
25            res[i] = lst + extend(i - R, i + R, N);
26            mid = i;
27            R = i + res[i] - 1;
28        } else {
29            res[i] = min(res[j], lst);
30        }
31    }
32    int mx = -1, idx = -1;
33    for (int i = 0; i < N; i++) {

```

```

34     if (res[i] > mx) {
35         mx = res[i];
36         idx = i;
37     }
38 }
39 cout << "Longest Palindromic Substring Length is " << (mx +
40     1) / 2 << "\n";
41 for (int i = idx - res[idx] + 1; i <= idx + res[idx] + 1; i
42     ++){
43     if (i & 1)
44         cout << s[i];
45     cout << "\n";
46 }

```

9.4 string trie

```

1 struct node {
2     node* link[26] = {nullptr};
3     bool flag = false;
4     bool contain_key(char ch) { return (link[ch - 'a'] != nullptr
5         ); }
6     void put(char ch, node* _node) { link[ch - 'a'] = _node; }
7     node* get(char ch) { return link[ch - 'a']; }
8     void set_end() { flag = true; }
9     bool is_end() { return flag; }
10 };
11 struct Trie {
12     node* root;
13     Trie() { root = new node(); }
14     void insert(string s) { // insert a string
15         node* now = root;
16         for (int i = 0; i < (int)s.size(); i++) {
17             if (!now->contain_key(s[i])) now->put(s[i], new node());
18             now = now->get(s[i]);
19         }
20         now->set_end();
21     }
22     // return if there is a word of the given string
23     bool query(string s) {
24         node* now = root;
25         for (int i = 0; i < (int)s.size(); i++) {
26             if (!now->contain_key(s[i])) return false;
27             now = now->get(s[i]);
28         }
29         return now->is_end();
30     }
31     // return if there is any word that start with the given
32     // prefix
33     bool query_prefix(string s) {
34         node* now = root;
35         for (int i = 0; i < (int)s.size(); i++) {
36             if (!now->contain_key(s[i])) return false;
37             now = now->get(s[i]);
38         }
39         return true;
40     }
41 };

```

9.5 xor trie

```

1 struct node {
2     node* link[2];
3     bool contain_key(int val) { return (link[val] != nullptr); }
4     node* get(int val) { return link[val]; }
5     void put(int val, node* _node) { link[val] = _node; }
6 };
7 struct Trie {
8     node* root;
9     Trie() { root = new node(); }
10    void update(int x) {
11        node* now = root;
12        for (int i = 30; i >= 0; i--) {
13            if (x & (1 << i)) {
14                if (!now->contain_key(1)) now->put(1, new node());
15                now = now->get(1);
16            } else {
17                if (!now->contain_key(0)) now->put(0, new node());
18                now = now->get(0);
19            }
20        }
21    }
22    int query(int x) { // query max xor with number x
23        node* now = root;
24        int mx = 0;
25        for (int i = 30; i >= 0; i--) {
26            int nowbit = (x >> i) & 1;
27            int target = nowbit ^ 1;
28            if (now->contain_key(target)) {
29                mx += (1 << i);
30                now = now->get(target);
31            } else {

```

```

32                now = now->get(nowbit);
33            }
34        }
35        return mx;
36    }
37 };

```

9.6 z

```

1 vector<int> Z;
2 void calculate_Z(string &s) {
3     Z.clear();
4     Z.resize(s.size(), 0);
5     int l = 0, r = 0;
6     for (int i = 1; i < s.size(); i++) {
7         if (i <= r)
8             Z[i] = min(Z[i - 1], r - i + 1);
9         while (i + Z[i] < s.size() && s[i + Z[i]] == s[Z[i]])
10             Z[i]++;
11         if (i + Z[i] - 1 > r) {
12             l = i; r = i + Z[i] - 1;
13         }
14     }
15 }
16 void Z_matching(string &a, string &b) {
17     string res = "";
18     int i = 0;
19     for (char ch : b)
20         res[i++] = ch;
21     res[i++] = '_';
22     for (char ch : a)
23         res[i++] = ch;
24     calculate_Z(res);
25     for (int i = 0; i < Z.size(); i++) {
26         if (Z[i] == b.size())
27             cout << "found a matching start at " << i - b.size() - 1
28                 << "\n";
29     }
30 }

```

10 tree

10.1 Heavy Light Decomposition

```

1 const int N = 100005;
2 vector<int> G[N];
3 struct HLD {
4     vector<int> pa, sz, depth, mxson, topf, id;
5     int n, idcnt = 0;
6     HLD(int _n) : n(_n), pa(_n + 1), sz(_n + 1), depth(_n + 1),
7         mxson(_n + 1), topf(_n + 1), id(_n + 1) {}
8     void dfs1(int v = 1, int p = -1) {
9         pa[v] = p; sz[v] = 1; mxson[v] = 0;
10        depth[v] = (p == -1 ? 0 : depth[p] + 1);
11        for (int u : G[v]) {
12            if (u == p) continue;
13            dfs1(u, v);
14            sz[v] += sz[u];
15            if (sz[u] > sz[mxson[v]]) mxson[v] = u;
16        }
17    }
18    void dfs2(int v = 1, int top = 1) {
19        id[v] = ++idcnt;
20        topf[v] = top;
21        if (mxson[v]) dfs2(mxson[v], top);
22        for (int u : G[v]) {
23            if (u == mxson[v] || u == pa[v]) continue;
24            dfs2(u, u);
25        }
26    }
27    int path_query(int a, int b) {
28        int res = 0;
29        while (topf[a] != topf[b]) {
30            if (depth[topf[a]] < depth[topf[b]]) swap(a, b);
31            res = max(res, query(id[topf[a]], id[a])); // query : l =
32                id[topf[a]], r = id[a]
33            a = pa[topf[a]];
34        }
35        if (depth[a] < depth[b]) swap(a, b);
36        res = max(res, query(id[b], id[a])); // query : l = id[b],
37            r = id[a]
38        return res;
39    }
40 }

```